Per-track momentum re-calibration (κ look-up table) tested with K^0_S and Λ mass peaks in sPHENIX run3pp p+p data

X. Dong (dongxs) – draft prepared with Hermes

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Abstract

A per-track transverse-momentum re-calibration, given as a look-up table $\kappa(q, p_T, \eta, \varphi)$ derived on runs 79509–79516 (`MomentumScaleMap/run79509to79516_20261102`), is applied to the daughters of $K^0_S \to \pi^+\pi^-$ and $\Lambda \to p\pi^-$ (+c.c.) candidates reconstructed with KFParticle in 287 run3pp segments (32 runs, 284,648 events). The invariant masses are recomputed with and without the correction from the same candidates, and the momentum scale s and resolution $\sigma(p_T)/p_T = \sqrt{(ap_T)^2 + (b/\beta)^2}$ are fitted to the mass peaks in V0-p_T bins with templates from the data-driven fast simulation (DDSim). The sample is split into period A (runs 79509–79516, the range of the table; 95 segments) and period B (the other 24 runs, 192 segments, where the table is applied as is). **Results** (K^0_S fit; the joint K^0_S+Λ fit gives the same points): period A $a = 3.00 \to 1.50\%/(\mathrm{GeV}/c)$, $b = 1.60 \to 1.40\%$, $s = 0.991 \to 0.999$; period B $a = 4.00 \to 2.00\%/(\mathrm{GeV}/c)$, $b = 1.20 \to 1.40\%$, $s = 0.991 \to 0.998$ (without $\to$ with κ); all segments combined $a = 3.50 \to 1.75\%/(\mathrm{GeV}/c)$, $s = 0.991 \to 0.998$. The p_T-integrated K^0_S peak moves from 494.78 to 497.21 MeV/c^2 in period A (PDG 497.611). The correction removes nearly all of the 0.9% momentum-scale offset and halves the curvature term in both periods; period B keeps a larger curvature term. A tighter Λ selection (DIRA>0.9995 and larger daughter DCAs) raises the Λ S/B from 0.35 to 1.65 and reduces the error of the Λ peak position by a quarter; it is used for the Λ scale ($s_\Lambda = 0.999$ with κ), while the resolution stays determined by K^0_S. Statistical uncertainties only; the DDSim defaults are not changed. All 287 segments are used. Boxes: blue = choice made (with reason), red = caveat, orange = needs a decision.

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1 Overview

Purpose. The run3pp K^0_S mass peak sits about $3\,\mathrm{MeV}/c^2$ below the PDG value, and its width grows with p_T much faster than the full simulation predicts. In the DDSim momentum calibration (DDSim analysis note, Sec. "Main result") this was absorbed into a global scale $s = 0.991$ and a large curvature term a. A per-track correction table $\kappa(q, p_T, \eta, \varphi)$ has since been derived for runs 79509–79516. This note applies it to the V0 daughters, recomputes the masses and repeats the (a, b, s) fit, to answer three questions:

1. Does the correction remove the momentum-scale offset ($s \to 1$)?
2. How much of the fitted resolution was a charge-, η- and φ-dependent scale distortion?
3. Does a table derived on runs 79509–79516 also work for later runs (period B)?

Findings.

- Scale: s moves from 0.991 to 0.999 (period A) and from 0.991 to 0.998 (period B). The p_T-integrated K^0_S Gaussian peak moves from 494.78 to 497.21 MeV/c^2 (A) and from 494.57 to 497.04 MeV/c^2 (B); PDG 497.611 MeV/c^2.
- Resolution: the curvature term falls to 0.50 (A) and 0.50 (B) of its uncorrected value: $a = 3.00 \to 1.50$ and $4.00 \to 2.00\%/(\mathrm{GeV}/c)$. The constant term b changes little ($1.60 \to 1.40$, $1.20 \to 1.40\%$). For pions at $2\,\mathrm{GeV}/c$ the fitted σ/p_T is 6.2→3.3% (A) and 8.1→4.2% (B).
- Period B, outside the range of the table, improves as much as period A, but keeps a larger curvature term after the correction (2.00 vs $1.50\%/(\mathrm{GeV}/c)$). Without the correction the curvature term already grows with the run period (Sec. 9).
- K^0_S and Λ agree after the correction: the Λ χ^2 at the K^0_S best point is 6.1 (A) and 4.8 (B) above the Λ minimum (3.4 and 30.5 without κ).
- A tighter Λ selection (Sec. 8) raises the Λ S/B from 0.35 to 1.65 and reduces the error of the Λ peak position from 0.076 to $0.056\,\mathrm{MeV}/c^2$; the Λ-only scale with κ is 0.999 (A) and 1.000 (B), consistent with K^0_S. Its Λ resolution agrees with K^0_S in period A but not in period B, so it is used for the scale only.
- Reproducibility and validation checks all pass (Sec. 10).

Choices (one line each).

- Same candidates, three mass versions (KFP fitted mass; recomputed without κ; recomputed with κ): the effect of the correction is "no κ" vs "with κ".
- Two run periods fitted separately: A = range of the table, B = all other runs (table applied as is, per user decision).
- No covariance- or significance-based cut anywhere (DDSim cannot reproduce covariance correlations).
- Fit model, templates, bins and background as in the DDSim main calibration; only the scale grid is widened.
- Λ scale from a tighter Λ selection, applied identically to data and templates; resolution from K^0_S.

2 Data sample, software and processing

Data. sPHENIX run3pp (p+p, $\sqrt{s}$ = 200 GeV, streaming readout), production `ana581_2026p005_v001`, `DST_TRKR_TRACKS` (tracks, track seeds, vertices; one DST per segment of about 990 events), from `/sphenix/lustre01/sphnxpro/production/run3pp/physics/ana581_2026p005_v001/DST_TRKR_TRACKS/`. 287 segments from 32 runs (79509–81609):

- run 79509 segments 0–30 (the sample of the first calibration);
- the segments of the TrackQA single-track scan: 8 segments per run for all 32 runs (run 79509: segments 220, 680, 1140, 1600, 2060, 2520, 2980, 3440).

The same segment list was used for the covariance-free production of the DDSim main calibration (condor cluster 480608).

Run periods.

- **Period A**: runs 79509–79516 (8 runs, 95 segments, 94,173 events) = the run range from which the κ table was derived (directory name `run79509to79516_20261102`).
- **Period B**: runs 80904–81609 (24 runs, 192 segments, 190,475 events). The table is applied unchanged, assuming it holds there (user decision).

Per-run counts are listed in Table 16 (Sec. 9). All 287 segments are used.

Software. Release `ana.581` (`sphenix_setup.sh -n ana.581` via `run3pp/setup_env.sh`), reconstruction macro derived from `macros/TrackingProduction/Fun4All_KShortReco.C` at macros commit `87f5cb5e` (the commit of ana.581): `run3pp/macro/loosecov/Fun4All_run3pp_KFP.C`. Conditions: CDB global tag `newcdbtag`, timestamp = run number, Acts tracking geometry from the CDB, MVTX misalignment applied, `TRACKING::streaming_mode = true`, TPC readout configuration per run (`TpcReadoutInit`). Mass-histogram step: `run3pp/macro/kappa/fill_v0_mass_kappa.C` (ROOT, interpreted). Fit and figures: DataDrivenSim (`macros/fitMomRes.C`, this note's `scripts/`).

Processing (condor cluster 480714). One job per segment (`run3pp/macro/run_v0mass_kappa.sh`, `run3pp/condor/v0mass_kappa.sub`, 3 GB, scratch on the worker):

1. KFParticle K^0_S and Λ reconstruction over all events of the segment (`macro/run_kfp_loosecov.sh`, cut set "loose" without covariance cuts, Sec. 3); the ntuples stay in the job scratch directory;
2. `fill_v0_mass_kappa.C`: event/track/candidate selection (Sec. 4) and three sets of mass-vs-p_T histograms, `V0mass_<base>_{kfp,raw,kappa}.root` (Sec. 5); a job is accepted only if both steps print "All done" and all three files are non-empty;
3. only the histograms (≈80 kB per job) and the gzipped logs are copied to `run3pp/output/v0mass_kappa`; the ntuples are deleted.

Result: 287/287 jobs OK, 0 failed, 0 resubmitted. A second pass over the same segments (condor 481293) added the tight Λ histograms and slim candidate trees (Sec. 8). Merging per period: `scripts/merge_kappa287.sh` → `data/kappa287/V0mass_{A,B,all}_{kfp,raw,kappa}.root`.

3 V0 reconstruction (KFParticle)

K^0_S and Λ candidates are built by `KFParticle_sPHENIX` from the `SvtxTrackMap` tracks and the `SvtxVertexMap` vertices of each event, one module per channel, with the settings of Table 1 (`run3pp/macro/loosecov/Fun4All_run3pp_KFP.C`, cut set "loose", the two covariance cuts removed). Decay descriptors: `K_S0 -> pi^+ pi^-` and `[Lambda0 -> proton^+ pi^-]cc`. All other topological requirements are switched off in KFParticle and applied offline (Sec. 4).

Table 1: KFParticle reconstruction settings (pre-cuts) for K^0_S and Λ, identical for both channels except the mass window. "off" = set to $\pm$`FLT_MAX` or -1 in the macro.

level	setting	value
event / vertex	primary vertices	`SvtxVertexMap` (`dontUseGlobalVertex(true)`)
	PV choice	`constrainToPrimaryVertex(true)`: candidate kept if at least one PV of the matching crossing passes the PV-dependent cuts; the PV is stored
	crossing (streaming readout)	both daughters from the same crossing (`requireBunchCrossingMatch(true)`); PV from that crossing (`requireTrackVertex-BunchCrossingMatch(true)`)
track	transverse momentum	$p_T > 0.1\,\mathrm{GeV}/c$
	TPC	≥ 20 TPC hits (`setMinTPChits(20)`)
	MVTX, INTT	no requirement (`setMinMVTXhits(0)`, `setMinINTThits(0)`)
	track χ^2/ndf	off
	track–PV DCA, DCA significance	off (`setMinimumTrackPV_DCA(-1)`, `..._StdDev(-1)`)
	PID	none (`usePID(false)`; dE/dx stored only)
pair / vertex	daughter–daughter DCA	< 0.5 cm (`setMaximumDaughterDCA`); DCA$_{xy}$ off
	SV χ^2/ndf	off (`FLT_MAX`; was < 20 in the original production)
	SV error-ellipsoid volume	off (`FLT_MAX`; was < 0.1 cm^3)
mother	pointing (DIRA $= \cos\theta$)	> 0.8
	mass window	K^0_S 0.40–0.60, Λ 1.05–1.20 GeV/c^2
	mother p_T	> 0
	flight distance χ^2, decay length (3D, xy), decay time, their significances, mother–PV DCA (and significance)	off

Implicit behaviour (from the ana.581 KFParticle source, read for the earlier run3pp studies).

- The track–PV DCA requirement, even at -1, needs a vertex in the track's own crossing: a silicon-matched track whose crossing has no vertex gets a NaN DCA and is dropped. TPC-only tracks carry crossing 0 by default.
- The daughter–daughter DCA used by `setMaximumDaughterDCA` is computed after both tracks are constrained to a common production point, so the 0.5 cm requirement rarely rejects a pair.
- With `[...]cc` one mass hypothesis is kept per track pair (Λ or $\bar{\Lambda}$).

Stored quantities used here (`DecayTree`). Mother: `<M>_mass`, `<M>_pT` (unconstrained secondary-vertex fit; mass not constrained), `<M>_DIRA`, `<M>_decayLength_xy` (with respect

to the chosen PV). Daughters: `track_N_px,py,pz` (momentum after transport to the fitted secondary vertex), `track_N_charge`, `track_N_pseudorapidity`, `track_N_PV_DCA_xy`; pair: `track_1_track_2_DCA` (after both daughters are constrained to the secondary vertex; about 20 μm, so the offline cut on it is ineffective); event: `primary_vertex_z`. Daughter order in all run3pp candidates: K^0_S `track_1` $= \pi^-$, `track_2` $= \pi^+$; Λ `track_2` = (anti)proton (`|track_2_PDG_ID| = 2212`). The stored `track_N_mass` is the fitted KF mass and is not used for species assignment.

Caveat

The sign of `Lambda0_PDG_ID` does not separate Λ and $\bar{\Lambda}$ in these ntuples. In this note Λ and $\bar{\Lambda}$ are separated by the charge of `track_2` (the proton), in all three mass versions, and fitted together.

4 Offline selection

The offline selection is the run3pp V0 QA selection used in the DDSim calibration. It is applied in `fill_v0_mass_kappa.C` with the stored KFParticle values, identically for the three mass versions (the cut variables are not recomputed after the momentum correction). Table 2 lists every level, including the requirements that are deliberately absent, and the corresponding DDSim template selection (Sec. 6).

Table 2: Offline selection of the calibration sample (data) and of the DDSim templates. "–" = deliberately no cut. The "loose" histograms (`h*_loose`) contain the event and track rows only; all fits use the QA histograms (all rows). The tight Λ selection of Sec. 8 adds DIRA>0.9995, $|\mathrm{DCA}_{xy}(\pi)| > 0.10$ cm and $|\mathrm{DCA}_{xy}(p)| > 0.03$ cm, in data and templates.

level	data (offline, on the KFParticle candidates)	DDSim templates
event	$\lvert z_{\mathrm{PV}}\rvert < 10$ cm (chosen PV); no trigger selection; no multiplicity or pile-up cut; no event de-duplication across overlapping streaming events (`Collision_BC0`)	generated PV $\lvert z\rvert < 10$ cm, (x, y) and z from data
track	$\lvert\eta\rvert < 1.1$ for both daughters; $\lvert\mathrm{DCA}_{xy}\rvert > 0.02$ cm to the chosen PV for both daughters; KFP pre-cuts of Table 1 ($p_T > 0.1$ GeV/c, TPC≥ 20)	$p_T^{\mathrm{rec}} > 0.1$ GeV/c, $\lvert\eta\rvert < 1.1$, $\lvert\mathrm{DCA}_{xy}\rvert > 0.02$ cm; TPC efficiency (full sim); MVTX≥ 2 drawn from the data matching efficiency; no silicon beyond $R_{xy} = 3.2$ cm
pair / vertex	$\mathrm{DCA}_{12} < 0.2$ cm (SV-constrained, ineffective); no vertex-χ^2 or volume cut	$\mathrm{DCA}_{12} < 0.2$ cm (unconstrained helices); no covariance
mother	DIRA> 0.995; $L_{xy} > 0.5$ cm; no upper L_{xy} cut; no mother-DCA or significance cut	$\cos\theta > 0.995$, $L_{xy} > 0.5$ cm
species	K^0_S: $\pi^+\pi^-$; Λ: `track_2` = (anti)proton, $\bar{\Lambda}$ if its charge is negative	K^0_S, Λ (50% $\bar{\Lambda}$)
candidates	all candidates kept (no best-candidate choice; a track can enter several candidates)	one candidate per decay

Histograms. Per mass version: `hKs_qa` (p_T 0–5 GeV/c in 100 bins $\times$ M 0.40–0.60 GeV/c^2 in 200 bins of 1 MeV/c^2), `hLa_qa`, `hLaBar_qa` (M 1.08–1.16 GeV/c^2 in 320 bins of 0.25 MeV/c^2), the same without the QA rows (`*_loose`), and the number of processed events (`nEventsProcessed` from the KFParticle `EventInfo` tree). The V0 p_T on the horizontal axis is the KFParticle p_T (KFP version) or the recomputed p_T (other versions).

Statistics. QA candidates (all masses in the histogram range, signal plus background): 19,841 K^0_S and 38,100 Λ+$\bar{\Lambda}$ in period A, 107,418 and 243,169 in period B. Table 3 gives the entries in the fit range per V0-p_T bin.

Table 3: Data entries (signal plus background) in the fit range per V0-p_T bin, KFP mass and κ-corrected versions (the recomputed p_T moves candidates between bins). From the fit logs. The recomputed versions have fewer entries in the fit window than the KFP version although the candidates are the same; the origin of this difference is part of open issue 3 (Sec. 11).

	V0 p_T (GeV/c)	A KFP	A with κ	B KFP	B with κ
K^0_S	0.2–0.5	3,601	3,028	16,997	13,422
	0.5–0.7	3,076	2,558	14,712	12,186
	0.7–0.9	2,497	2,049	12,327	9,169
	0.9–1.2	2,286	1,750	10,882	7,138
	1.2–1.6	1,216	1,039	5,576	4,028
	1.6–2.2	561	514	2,281	1,902
	2.2–4.0	225	201	952	755
$\Lambda+\bar{\Lambda}$	0.5–0.9	4,686	3,726	30,258	23,394
	0.9–1.2	1,652	1,178	9,865	6,516
	1.2–1.6	835	687	4,627	3,387
	1.6–2.4	350	336	1,620	1,409
	2.4–4.0	74	73	357	281

Caveat

The DDSim templates were generated with $\mathrm{DCA}_{12} < 0.2$ cm on unconstrained helices, while the effective data pair cut is KFParticle's 0.5 cm on constrained tracks (Sec. 3). The DDSim defaults have since been changed to 0.5 cm; the templates of this note (and of the DDSim main calibration) predate that change. The cut changes which decays enter the template, not the momentum response; its effect on (a, b, s) is not evaluated.

5 The κ correction and the three mass versions

Table. `MomentumScaleMap/run79509to79516_20261102/kappa_lookup.csv`, frozen on the cluster as `run3pp/macro/kappa/kappa_lookup_run79509to79516_20261102.csv` (md5 `38e0a20c...`). Columns `q, pT, curv, eta, phi, kappa, kappa_curv, eps, delta`; 72,800 nodes = 2 charges × 112 p_T values (0.1–20 GeV/c, 0.05 GeV/c steps up to 5 GeV/c, coarser above) × 13 η values (-1.5 to 1.5, step 0.25) × 25 φ values ($-\pi$ to π, step $\pi/12$; both ends present and equal). Convention (file header): $(1/p_T)^{\mathrm{corr}} = (1/p_T)\,\kappa_{\mathrm{curv}}$ with $\kappa_{\mathrm{curv}} = 1 + \epsilon + q\,\delta$, and $\kappa = 1/\kappa_{\mathrm{curv}}$ is the multiplier of the transverse momentum, $p_T^{\mathrm{corr}} = \kappa\, p_T$. According to the file header the correction is parametrised in $x = \ln p_T$ (reference -0.588194, scale 0.790816, clamp 6.0; κ constant in x outside the clamp). ϵ is charge-even (a scale), $q\delta$ charge-odd (a curvature offset, i.e. a sagitta-like distortion).

Table 4: Table values κ at $\eta = 0$, $\varphi = 0$ (nodes, no interpolation). Over the whole table κ ranges from 0.646 to 1.424 (extremes at large p_T and $|\eta|$).

p_T (GeV/c)	$\kappa(q=-1)$	$\kappa(q=+1)$
0.1	1.0006	1.0103
0.2	0.9984	1.0074
0.3	0.9952	1.0115
0.5	0.9890	1.0211
0.7	0.9827	1.0289
1	0.9725	1.0378
1.5	0.9549	1.0481
2	0.9379	1.0555
3	0.9074	1.0666
5	0.8597	1.0838

The charge-odd part dominates above $\approx 0.5\,\mathrm{GeV}/c$: at $\eta = \varphi = 0$, $\kappa(q=-1)$ falls and $\kappa(q=+1)$ rises with p_T. The table also varies with φ (at $p_T = 0.5\,\mathrm{GeV}/c$, $\eta = 0$: 0.986–1.015 for $q=-1$, 0.988–1.023 for $q=+1$) and with η (0.964–0.997 and 1.018–1.068 at $\varphi = 0$).

Application. For each daughter of each candidate, with the stored momentum (at the fitted secondary vertex) and charge:

1. κ is looked up by linear interpolation in $(\ln p_T, \eta, \varphi)$ between the eight surrounding nodes, separately for $q = \pm 1$; outside the node range the value is clamped to the edge ($p_T < 0.1$ or $> 20\,\mathrm{GeV}/c$, $|\eta| > 1.5$);
2. all three momentum components are multiplied by κ (p_T scaled, η and φ unchanged);
3. the V0 mass is recomputed from the two daughter four-vectors with PDG masses ($m_\pi = 139.570$, $m_p = 938.272\,\mathrm{MeV}/c^2$), and the V0 p_T from the sum of the daughter momenta.

Mass versions. From the same candidates and the same selection, each job writes

- **KFP**: the KFParticle mass and p_T (identical to the earlier production 480608, Sec. 10);
- **no κ** ("raw"): mass and p_T recomputed from the stored daughter momenta, $\kappa \equiv 1$;
- **with κ**: as "raw" after the correction.

Choice

The effect of the correction is measured as "with κ" vs "no κ": same candidates, same mass formula, only the daughter momenta differ. On run 79509 segments 0–30, recomputed minus KFP mass is +0.14 (RMS 4.5) MeV/c^2 for K^0_S and +0.44 (RMS 3.9) MeV/c^2 for Λ, all candidates; the origin of the difference (KFParticle vertex fit vs stored momenta) is not investigated.

Mean correction applied. QA-selected candidates, averaged over all masses in the histogram range (job logs):

- period A: K^0_S π^- 0.9979, π^+ 1.0163; Λ pion 1.0077, proton 1.0065;
- period B: K^0_S π^- 0.9977, π^+ 1.0162; Λ pion 1.0077, proton 1.0066.

The p_T dependence on run 79509 segments 0–30 is in Table 5.

Table 5: Mean κ applied per daughter in $0.1\,\mathrm{GeV}/c$ bins of daughter p_T (bin containing the quoted value), all candidates of run 79509 segments 0–30 before the QA rows (local study, `macros/fillV0MassCorr.C`); "–" fewer than 50 daughters. For Λ the pion and proton columns mix both charges (Λ and $\bar{\Lambda}$).

daughter	0.15	0.3	0.5	0.8	1.2	2	3
K^0_S: π^-	1.004	1.002	0.998	0.989	0.976	0.950	–
K^0_S: π^+	1.010	1.012	1.019	1.027	1.035	1.048	–
Λ+$\bar{\Lambda}$: π	1.007	1.007	1.012	1.019	1.015	1.029	–
Λ+$\bar{\Lambda}$: p, $\bar{\mathrm{p}}$	1.008	1.008	1.008	1.008	1.004	0.992	–

Caveat

The table header states that κ was trained on vertex-constrained daughter kinematics and deployed on raw `SvtxTrackMap` tracks. Here it is applied to KFParticle daughter momenta constrained to the secondary vertex. Whether this is the same constraint as in the training is not verified.

6 Momentum scale and resolution fit

The method is that of the DDSim main calibration (`macros/fitMomRes.C`); it is summarised here with every setting.

Response model. Each DDSim daughter momentum is rescaled as

$$p^{\mathrm{rec}} = s\,p\,[1+\delta_{sp}(p_T)+\sigma(p_T)\,G], \qquad \frac{\sigma(p_T)}{p_T} = \sqrt{(a\,p_T)^2+(b/\beta)^2}, \qquad \beta = p/E, \quad G\sim\mathcal{N}(0,1), \tag{1}$$

direction unchanged. s is a global momentum scale, a the curvature (point-resolution) term, b the multiple-scattering term; the $1/\beta$ factor lets one (a,b) describe pions and protons. δ_{sp} is the mean momentum bias of the full simulation, non-zero only for protons (`hMomShift_p` of the DDSim input, energy-loss correction with the pion mass), and is applied in the templates of all three mass versions. The κ correction acts on the data only; the templates are the same for all versions.

Templates. DDSim (`output/momfit/{Ks,Lambda}_nosmear_v2.root`, 6M decays each, generated once):

- input `input/ddsim_input_run3pp_v2.root`: PV distribution, DCA maps and silicon-matching efficiencies from the run3pp TrackQA ntuples (356 segments); TPC efficiency and δ_{sp} from the full simulation (sPHENIX_202611, ana.582);
- generation: V0 p_T flat in 0–5 GeV/c and weighted to the STAR p+p 200 GeV K^0_S, Λ, $\bar\Lambda$ Lévy spectra (`wLevy`), $|y|<1$, $|z_{\mathrm{PV}}|<10$ cm, PDG $c\tau$, Λ 50% $\bar\Lambda$, decays isotropic in the rest frame;
- reconstruction: daughter positions smeared with the data DCA maps, momenta *not* smeared (`-no-mom-smear`), helix propagation in 1.4 T, secondary vertex at the closest approach of the two helices, true PV ($\sigma_{\mathrm{PV}}=0$);
- selection: `passAll` = acceptance + TPC efficiency + MVTX$\geq$ 2 for both daughters + the topology cuts of Table 2; 561,400 K^0_S and 255,316 Λ candidates pass;
- the daughter momenta at the reconstructed secondary vertex are stored, and for each candidate two Gaussian numbers G_1, G_2 are drawn once (fixed seeds) and reused at every grid point, so the χ^2 surface is smooth.

For a parameter point (a,b,s) the stored momenta are rescaled with Eq. 1, the mass and V0 p_T recomputed, and the weighted candidates filled into one mass template per V0-p_T bin. The selection is applied before the rescaling.

Fit in V0-p_T bins.

- K^0_S: V0 p_T bins 0.2, 0.5, 0.7, 0.9, 1.2, 1.6, 2.2, 4.0 GeV/c; mass 0.44–0.56 GeV/c^2 in 2 MeV/c^2 bins.
- Λ+$\bar\Lambda$: V0 p_T bins 0.5, 0.9, 1.2, 1.6, 2.4, 4.0 GeV/c; mass 1.103–1.130 GeV/c^2 in 0.5 MeV/c^2 bins (narrow window; the background under the Λ peak is large and curved at low p_T).
- Model per bin: $N\times$template + quadratic polynomial in M (4 free parameters), Poisson maximum likelihood by iteratively re-weighted least squares (variance = model); goodness of fit = Baker–Cousins likelihood-ratio χ^2, summed over the p_T bins. Template statistical errors are neglected (561,400 K^0_S template candidates against 77,189 data entries in the fit range of the full sample, 255,316 against 54,324 for Λ).
- Grid: a = 0–0.08 (step 0.0025) (GeV/c)$^{-1}$, b = 0.004–0.040 (step 0.002), s = 0.975–1.015 (step 0.001; widened from 0.984–1.000 of the main calibration because the correction moves s to ≈ 1). Best point = grid minimum.

- Uncertainties: profile $\Delta\chi^2 = 1$ in each parameter (minimum over the other two), interpolated linearly between grid points; statistical only. A grid-edge flag is printed per fit.
- K^0_S and Λ are fitted separately and jointly (sum of the two χ^2); the Λ χ^2 at the K^0_S best point measures their compatibility.

Each fit (one per period and mass version) is one call of `scripts/run_momfit_kappa.sh`.

Caveat

The parameters are effective: (a, b) absorb every source of mass width that DDSim does not model separately, in particular the daughter angular resolution (directions are not smeared), residual misalignment and TPC distortions, and differences between the KFParticle vertex-fit mass and the recomputed mass. Below $\approx 1\,\mathrm{GeV}/c$ daughter p_T the two terms are strongly anti-correlated.

7 Results

7.1 Summary

Table 6: Best (a, b, s) of the K^0_S fit per run period and mass version (the joint $K^0_S{+}\Lambda$ fit gives the same point where the last column says "yes"), K^0_S Baker–Cousins χ^2/ndf, and the Λ χ^2 at the K^0_S best point minus the Λ minimum ($\Delta\chi^2_\Lambda$). Uncertainties and the Λ-only and joint fits: Table 7.

period	mass	a (%/(GeV/c))	b (%)	s	χ^2_K/ndf	$\Delta\chi^2_\Lambda$	joint = K^0_S
A	KFP mass	3.00	1.60	0.991	470.4/389	2.2	yes
	recomputed, no κ	3.00	1.60	0.991	464.5/389	3.4	yes
	recomputed, with κ	1.50	1.40	0.999	468.5/389	6.1	yes
B	KFP mass	4.00	1.20	0.991	400.5/389	13.8	no
	recomputed, no κ	4.00	1.20	0.991	473.1/389	30.5	yes
	recomputed, with κ	2.00	1.40	0.998	457.4/389	4.8	yes
all	KFP mass	3.75	1.40	0.991	421.5/389	13.9	yes
	recomputed, no κ	3.50	1.40	0.991	462.1/389	25.0	yes
	recomputed, with κ	1.75	1.40	0.998	473.9/389	9.0	no

Effect of the correction ("no κ" → "with κ", same candidates):

- **Scale.** $s = 0.991 \to 0.999$ (A) and $0.991 \to 0.998$ (B). The Λ-only fits agree with κ ($s = 0.998$ and 1.001). The p_T-integrated K^0_S Gaussian peak moves from 494.78 ± 0.10 to $497.21 \pm 0.07\,\mathrm{MeV}/c^2$ (A; -28.6σ and -5.5σ from PDG) and from 494.57 ± 0.11 to $497.04 \pm 0.08\,\mathrm{MeV}/c^2$ (B; -26.7σ, -6.9σ). With κ the K^0_S peak is still 0.40 (A) and $0.57\,\mathrm{MeV}/c^2$ (B) below PDG; the fitted $s < 1$ reflects the same residual.
- **Resolution.** a is reduced to 0.50 (A) and 0.50 (B) of its value without κ; b changes from 1.60 to 1.40% (A) and from 1.20 to 1.40% (B). The K^0_S Gaussian width at V0 $p_T = 1.6$–$2.2\,\mathrm{GeV}/c$ drops from 8.8 to $6.2\,\mathrm{MeV}/c^2$ (A) and from 9.0 to $6.3\,\mathrm{MeV}/c^2$ (B) (Table 8). Interpretation: without the correction the charge-odd and φ/η-dependent scale distortion that κ removes broadens the mass peak (π^+ and π^- are shifted in opposite directions, more at high p_T), and the fit absorbs this into a. This interpretation is not tested separately.
- **K^0_S/Λ consistency.** $\Delta\chi^2_\Lambda = 3.4 \to 6.1$ (A) and $30.5 \to 4.8$ (B). The K^0_S/Λ difference stays small in period A and largely disappears in period B with the correction.
- **Periods.** After the correction period B still has a larger curvature term than period A ($a = 2.00^{+0.05}_{-0.09}$ vs $1.50^{+0.02}_{-0.16}$%/(GeV/c)), as it had before (Sec. 9).

- **Fit quality.** With κ the K^0_S χ^2 is 468.5/389 (A) and 457.4/389 (B), against 464.5 and 473.1 without. In period B the KFP-mass fit has a clearly lower χ^2 (400.5) than both recomputed versions (KFP best point $a = 4.00\%/(\mathrm{GeV}/c)$, $b = 1.20\%$, $s = 0.991$; open issue).

Table 7: All fits: K^0_S, $\Lambda+\bar{\Lambda}$ and joint, per period and mass version. Profile $\Delta\chi^2 = 1$ intervals on the grid (statistical only); $\Delta\chi^2_\Lambda$ as in Table 6; "edge" = best point on the grid boundary. From `logs/fitMomRes_k287_<period>_<version>.txt`.

	mass	fit	a (%/(GeV/c))	b (%)	s	χ^2/ndf	$\Delta\chi^2_\Lambda$	edge
A	KFP mass	K^0_S	$3.00^{+0.06}_{-0.27}$	$1.60^{+0.03}_{-0.05}$	$0.9910^{+0.0001}_{-0.0001}$	470.4/389	2.2	no
		Λ	$1.75^{+0.72}_{-1.75}$	$2.00^{+0.37}_{-0.83}$	$0.9900^{+0.0040}_{-0.0017}$	261.8/247		no
		joint	$3.00^{+0.06}_{-0.27}$	$1.60^{+0.03}_{-0.03}$	$0.9910^{+0.0001}_{-0.0001}$	734.5/639		no
A	recomputed, no κ	K^0_S	$3.00^{+0.12}_{-0.13}$	$1.60^{+0.01}_{-0.10}$	$0.9910^{+0.0002}_{-0.0001}$	464.5/389	3.4	no
		Λ	$4.25^{+0.30}_{-0.18}$	$1.60^{+0.10}_{-0.22}$	$0.9910^{+0.0007}_{-0.0009}$	293.2/247		no
		joint	$3.00^{+0.05}_{-0.08}$	$1.60^{+0.02}_{-0.04}$	$0.9910^{+0.0002}_{-0.0001}$	761.1/639		no
A	recomputed, with κ	K^0_S	$1.50^{+0.02}_{-0.16}$	$1.40^{+0.01}_{-0.01}$	$0.9990^{+0.0000}_{-0.0001}$	468.5/389	6.1	no
		Λ	$3.25^{+0.60}_{-0.45}$	$1.20^{+0.08}_{-0.30}$	$0.9980^{+0.0022}_{-0.0004}$	252.7/247		no
		joint	$1.50^{+0.02}_{-0.08}$	$1.40^{+0.01}_{-0.01}$	$0.9990^{+0.0000}_{-0.0001}$	727.3/639		no
B	KFP mass	K^0_S	$4.00^{+0.29}_{-0.18}$	$1.20^{+0.15}_{-0.05}$	$0.9910^{+0.0001}_{-0.0008}$	400.5/389	13.8	no
		Λ	$6.00^{+0.37}_{-0.28}$	$1.80^{+0.18}_{-0.17}$	$0.9970^{+0.0026}_{-0.0010}$	276.2/247		no
		joint	$4.00^{+0.34}_{-0.28}$	$1.40^{+0.06}_{-0.23}$	$0.9910^{+0.0001}_{-0.0007}$	689.1/639		no
B	recomputed, no κ	K^0_S	$4.00^{+0.05}_{-0.07}$	$1.20^{+0.06}_{-0.04}$	$0.9910^{+0.0001}_{-0.0002}$	473.1/389	30.5	no
		Λ	$5.25^{+0.33}_{-0.29}$	$3.00^{+0.10}_{-0.11}$	$1.0050^{+0.0009}_{-0.0017}$	287.9/247		no
		joint	$4.00^{+0.05}_{-0.36}$	$1.20^{+0.23}_{-0.02}$	$0.9910^{+0.0001}_{-0.0002}$	791.5/639		no
B	recomputed, with κ	K^0_S	$2.00^{+0.05}_{-0.09}$	$1.40^{+0.04}_{-0.02}$	$0.9980^{+0.0002}_{-0.0000}$	457.4/389	4.8	no
		Λ	$3.50^{+0.16}_{-0.72}$	$1.40^{+0.24}_{-0.13}$	$1.0010^{+0.0008}_{-0.0014}$	284.0/247		no
		joint	$2.00^{+0.06}_{-0.06}$	$1.40^{+0.03}_{-0.02}$	$0.9980^{+0.0002}_{-0.0000}$	746.2/639		no
all	KFP mass	K^0_S	$3.75^{+0.07}_{-0.27}$	$1.40^{+0.05}_{-0.05}$	$0.9910^{+0.0000}_{-0.0002}$	421.5/389	13.9	no
		Λ	$5.75^{+0.51}_{-0.14}$	$1.80^{+0.13}_{-0.17}$	$0.9960^{+0.0012}_{-0.0012}$	282.6/247		no
		joint	$3.75^{+0.04}_{-0.13}$	$1.40^{+0.19}_{-0.03}$	$0.9910^{+0.0001}_{-0.0002}$	718.0/639		no
all	recomputed, no κ	K^0_S	$3.50^{+0.04}_{-0.05}$	$1.40^{+0.02}_{-0.03}$	$0.9910^{+0.0001}_{-0.0001}$	462.1/389	25.0	no
		Λ	$5.25^{+0.52}_{-0.29}$	$2.00^{+0.53}_{-0.30}$	$0.9960^{+0.0025}_{-0.0013}$	280.5/247		no
		joint	$3.50^{+0.06}_{-0.05}$	$1.40^{+0.04}_{-0.02}$	$0.9910^{+0.0001}_{-0.0001}$	767.6/639		no
all	recomputed, with κ	K^0_S	$1.75^{+0.03}_{-0.03}$	$1.40^{+0.01}_{-0.01}$	$0.9980^{+0.0006}_{-0.0000}$	473.9/389	9.0	no
		Λ	$2.75^{+0.48}_{-0.63}$	$1.20^{+0.30}_{-0.46}$	$1.0000^{+0.0013}_{-0.0011}$	255.7/247		no
		joint	$1.75^{+0.03}_{-0.02}$	$1.40^{+0.01}_{-0.01}$	$0.9990^{+0.0000}_{-0.0007}$	737.1/639		no

Λ-only fits on the grid edge: none. K^0_S and joint fits: none (check PASS).

7.2 Single-track resolution

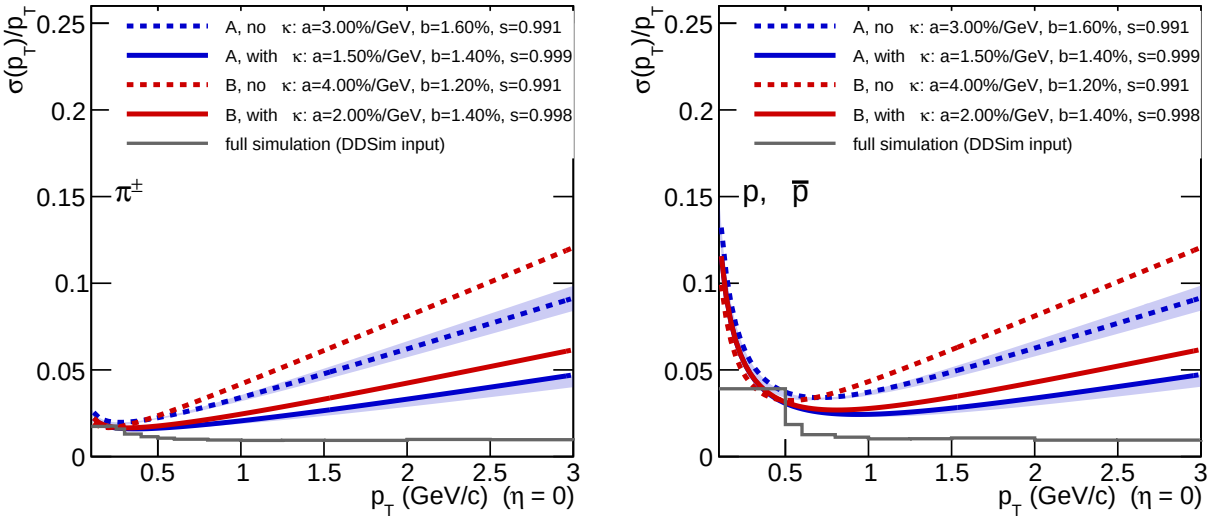


Figure 1: Fitted $\sigma(p_T)/p_T$ at $\eta = 0$ (Eq. 1) for pions (left) and protons (right): period A (blue) and B (red), without (dashed) and with (solid) κ; lines at the K^0_S best point (Table 6), bands = envelope of the (a, b) grid nodes with K^0_S $\Delta\chi^2 < 2.3$ (profiled over s; only the best node passes for B no κ, B with κ, so those bands are hidden under their lines). Grey: full-simulation response of the DDSim input. Script `scripts/plot_res.C`.

For pions at 1, 2, 3 GeV/c: period A 3.4, 6.2, 9.1% without and 2.1, 3.3, 4.7% with κ; period B 4.2, 8.1, 12.1% and 2.5, 4.2, 6.2%. Above ≈ 1.5 GeV/c the curves are constrained mainly by the two highest K^0_S bins (Table 3).

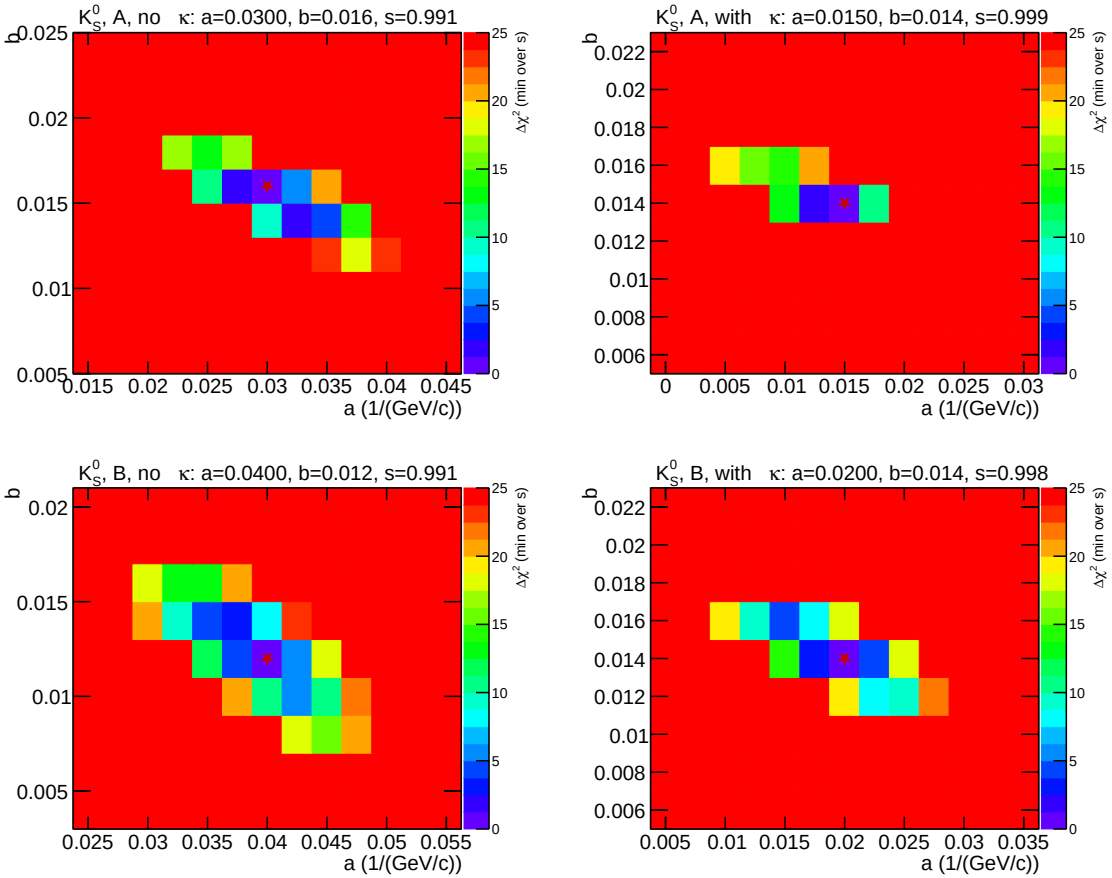


Figure 2: $\Delta\chi^2$ of the K^0_S fit in the (a, b) plane, minimised over s, on the grid nodes (cells; colour scale capped at 25): period A (top) and B (bottom), without (left) and with (right) κ. Red star: best node. The $\Delta\chi^2 < 2.3$ region is only a few nodes wide, so the grid step (0.0025 in a, 0.002 in b) limits the resolution of the intervals. Each panel shows ± 0.016 $(\mathrm{GeV}/c)^{-1}$ in a and ± 0.009 in b around its best node.

7.3 Invariant-mass distributions before and after the correction

Figure 3 compares the p_T-integrated mass distributions of the same QA-selected candidates, recomputed from the daughter momenta without κ (blue) and with κ (red); the KFParticle mass is shown for reference (grey). Only the daughter p_T changes; η, φ and the selection are the same. Figures 4–7 show the same per V0-p_T bin of the momentum fit (the V0 p_T is that of each version, so a candidate can change bin).

- K^0_S: with κ the peak moves up towards the PDG mass and becomes narrower and higher in both periods (Gaussian fit, Table 8: mean 494.78 → 497.21, σ 5.30 → 4.03 MeV/c^2 in A; 494.57 → 497.04, 5.43 → 4.33 MeV/c^2 in B). The change is largest above 1.2 GeV/c, where the uncorrected peak is broad and the corrected one narrow.
- Λ: mean 1115.55 → 1115.92, σ 2.16 → 1.63 MeV/c^2 (A) and 1116.24 → 1116.11, 2.69 → 1.88 MeV/c^2 (B), PDG 1115.683 MeV/c^2. The narrowing is smaller in absolute terms than for K^0_S, as expected from the small Λ decay momentum (the mass is less sensitive to the daughter momenta).
- The background is the same in both recomputed versions. The number of entries in the plotted mass range differs by less than 1% between them (K^0_S A 10,973 vs 10,974, B 47,446 vs 47,311), as expected since the correction only moves masses by a few MeV/c^2.
- The KFParticle mass has more combinatorial background in the same mass window (the recomputed spectra contain 0.83 (A) and 0.77 (B) of the KFP K^0_S entries, 0.78 and 0.75 for

Λ), although the candidates are the same. The cause is not investigated (open issue 3).

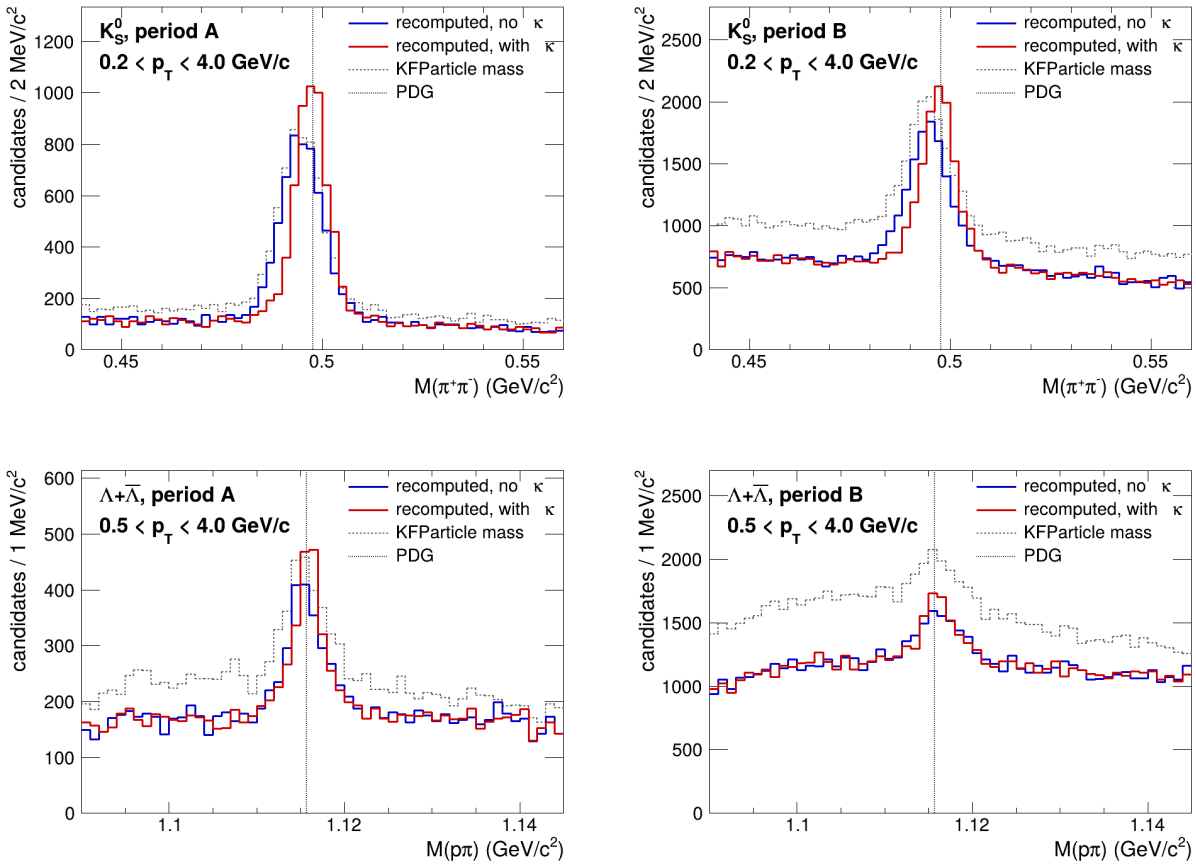


Figure 3: Invariant-mass distributions, p_T-integrated (K^0_S $0.2 < p_T < 4.0$, Λ+$\bar{\Lambda}$ $0.5 < p_T < 4.0\,\mathrm{GeV}/c$), period A (left) and B (right): recomputed without κ (blue), with κ (red), KFParticle mass (grey dashed); dotted line: PDG mass. Same QA-selected candidates in all curves. Script `scripts/massdist.C`.

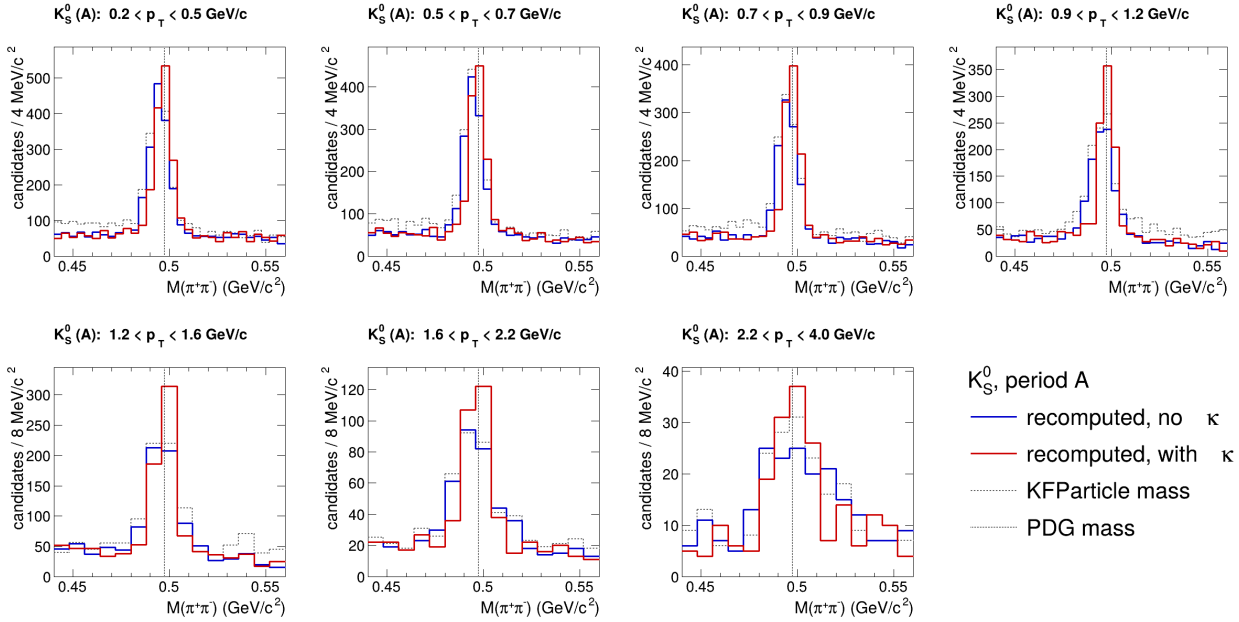


Figure 4: K^0_S mass per V0-p_T bin, period A, without (blue) and with (red) κ, KFParticle mass (grey dashed). Bin width 2, 4 or 8 MeV/c^2 depending on the statistics of the panel (axis title).

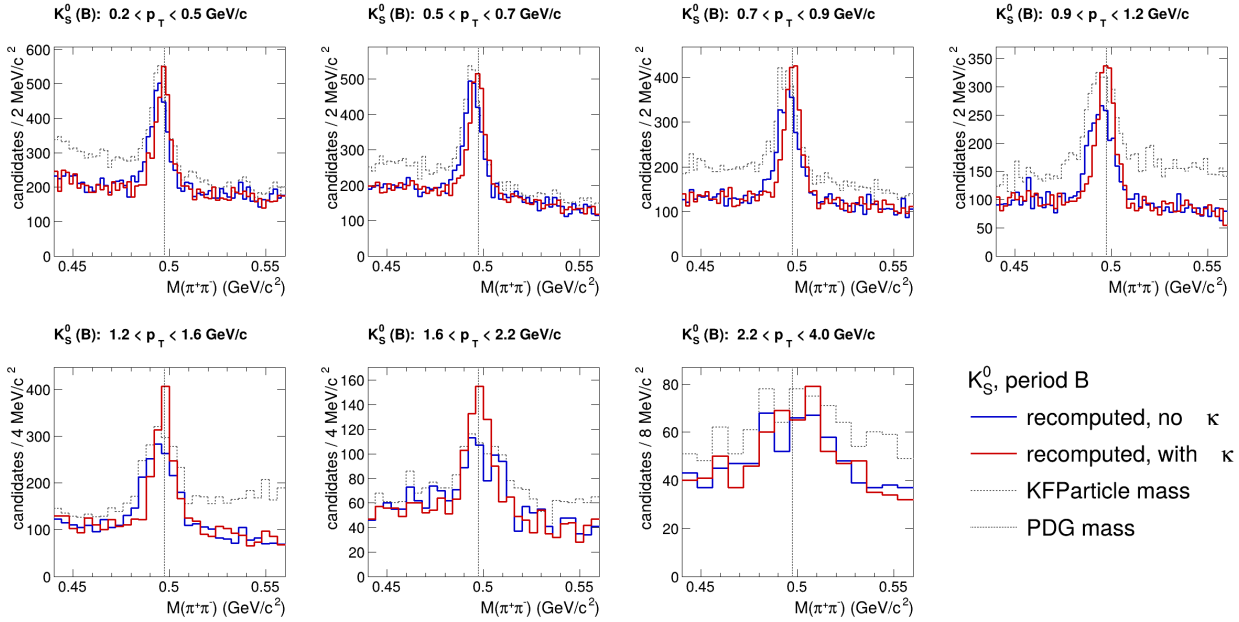


Figure 5: As Fig. 4, period B.

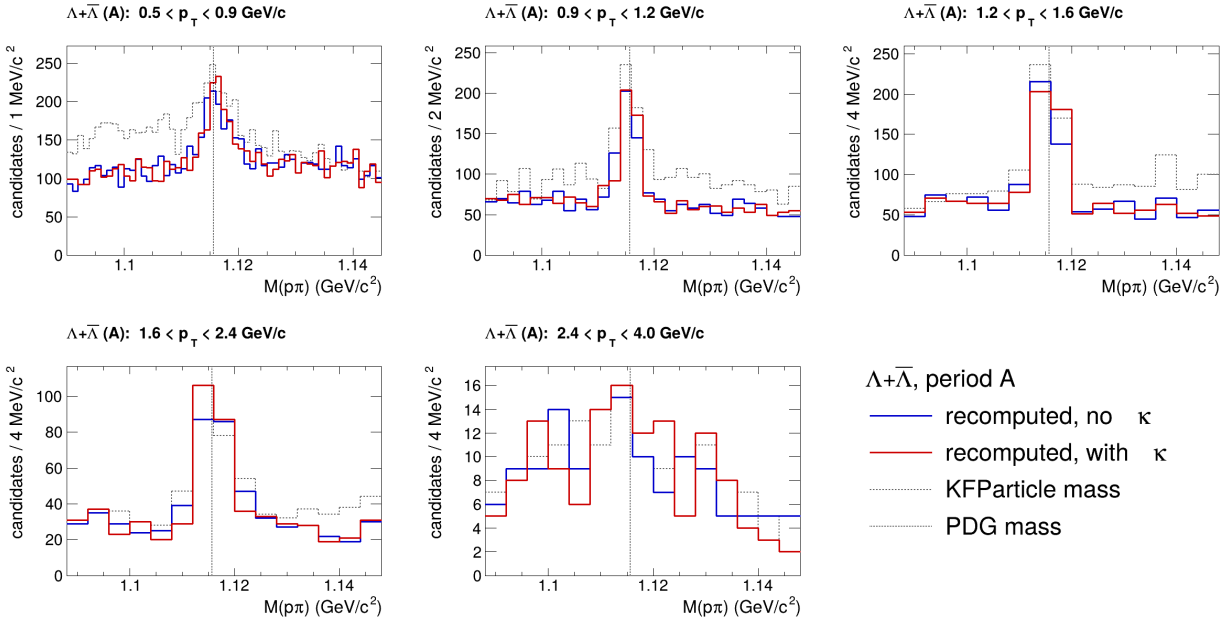


Figure 6: $\Lambda+\bar{\Lambda}$ mass per V0-p_T bin, period A, without (blue) and with (red) κ, KFParticle mass (grey dashed). Bin width 1, 2 or 4 MeV/c^2 depending on the statistics of the panel.

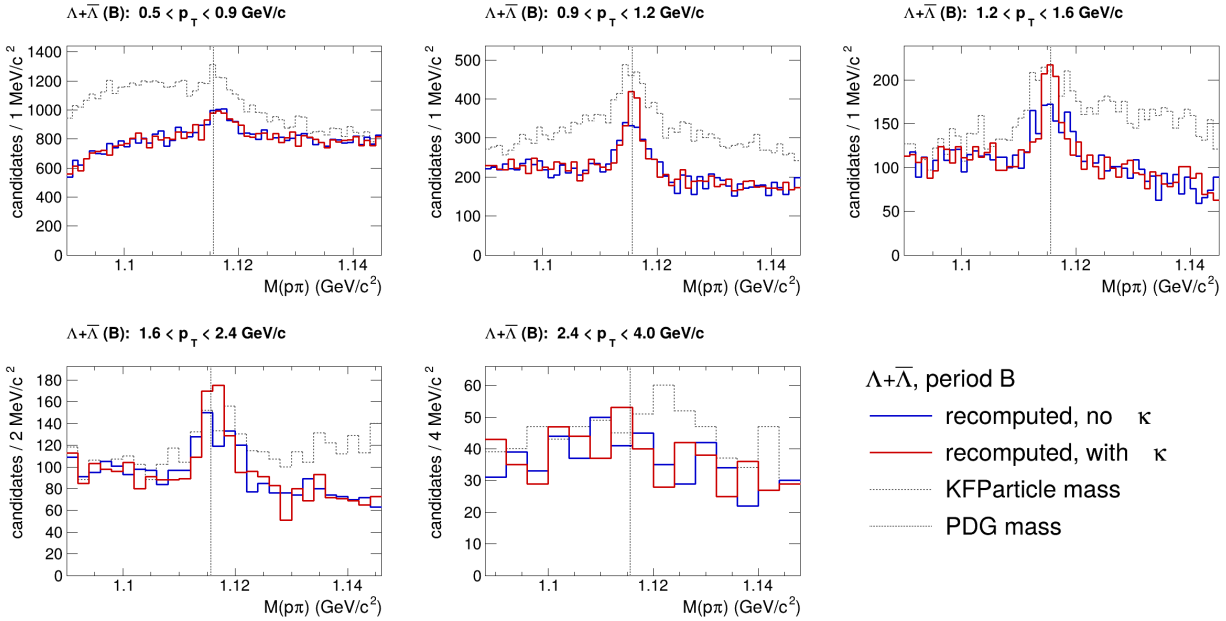


Figure 7: As Fig. 6, period B.

7.4 Mass peaks

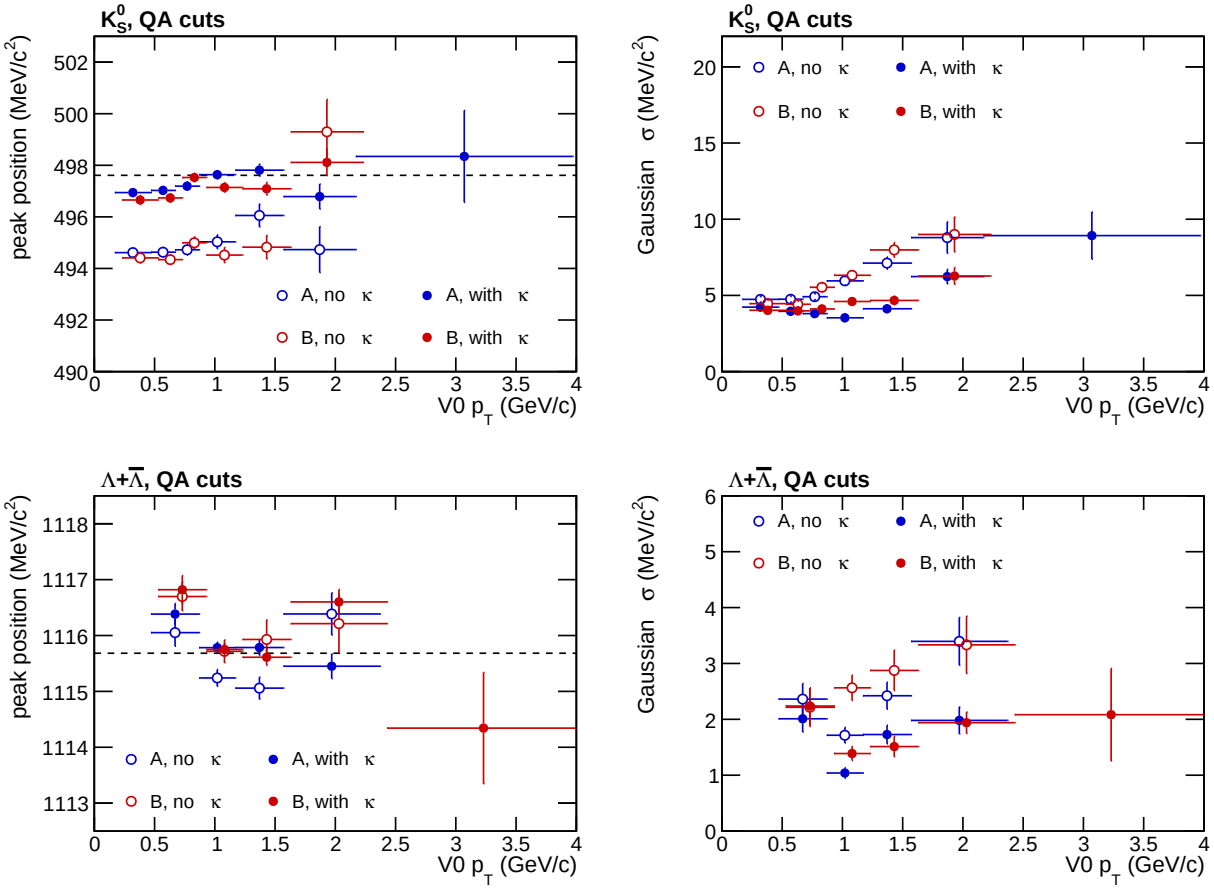


Figure 8: Gaussian peak position (left) and width (right) per V0-p_T bin for K^0_S (top; Gaussian + linear, 0.46–0.54 GeV/c^2) and Λ+$\bar{\Lambda}$ (bottom; Gaussian + quadratic, 1.103–1.130 GeV/c^2): period A (blue) and B (red), without (open) and with (full) κ; points of the two periods are offset by ± 0.03 GeV/c for visibility. Dashed: PDG mass. Unconstrained fits (mean error > 2 MeV/c^2 or width error $> 60\%$) are not drawn. Display only; the calibration uses the template fit. Script `scripts/peaks.C`.

Table 8: K^0_S Gaussian peak position M and width σ (MeV/c^2) per V0-p_T bin (GeV/c), fits of Fig. 8; "–" unconstrained. PDG 497.611 MeV/c^2.

	period A				period B			
p_T	M, no κ	σ, no κ	M, κ	σ, κ	M, no κ	σ, no κ	M, κ	σ, κ
0.2–0.5	494.61±0.17	4.74±0.16	496.94±0.15	4.23±0.15	494.41±0.20	4.46±0.22	496.65±0.18	4.02±0.20
0.5–0.7	494.63±0.18	4.75±0.17	497.02±0.15	3.96±0.15	494.34±0.19	4.42±0.20	496.73±0.16	3.98±0.17
0.7–0.9	494.72±0.20	4.91±0.18	497.19±0.15	3.80±0.13	494.99±0.24	5.53±0.27	497.53±0.17	4.11±0.17
0.9–1.2	495.03±0.27	5.95±0.27	497.64±0.15	3.53±0.14	494.52±0.30	6.32±0.30	497.14±0.19	4.60±0.19
1.2–1.6	496.05±0.45	7.12±0.39	497.81±0.23	4.12±0.23	494.82±0.46	7.99±0.49	497.09±0.25	4.67±0.28
1.6–2.2	494.73±0.89	8.79±1.03	496.79±0.48	6.24±0.48	499.30±1.25	9.00±1.15	498.11±0.51	6.27±0.56
2.2–4.0	–	–	498.34±1.78	8.92±1.55	–	–	–	–
0.2–4.0 (all)	494.78±0.10	5.30±0.10	497.21±0.07	4.03±0.07	494.57±0.11	5.43±0.13	497.04±0.08	4.33±0.09

Table 9: As Table 8 for $\Lambda+\bar{\Lambda}$. PDG 1115.683 MeV/c^2.

	period A				period B			
p_T	M, no κ	σ, no κ	M, κ	σ, κ	M, no κ	σ, no κ	M, κ	σ, κ
0.5–0.9	1116.05±0.24	2.36±0.28	1116.38±0.19	2.01±0.24	1116.70±0.26	2.21±0.35	1116.82±0.25	2.24±0.32
0.9–1.2	1115.24±0.15	1.72±0.14	1115.78±0.10	1.04±0.09	1115.72±0.20	2.56±0.23	1115.75±0.10	1.39±0.13
1.2–1.6	1115.06±0.20	2.42±0.24	1115.79±0.14	1.73±0.17	1115.93±0.35	2.88±0.36	1115.61±0.15	1.51±0.18
1.6–2.4	1116.39±0.38	3.40±0.43	1115.45±0.22	1.98±0.24	1116.21±0.52	3.33±0.51	1116.60±0.22	1.94±0.19
2.4–4.0	–	–	–	–	–	–	1114.34±1.00	2.08±0.83
0.5–4.0 (all)	1115.55±0.12	2.16±0.14	1115.92±0.08	1.63±0.11	1116.24±0.16	2.69±0.19	1116.11±0.10	1.88±0.13

7.5 Template fits per V0-p_T bin

Figures 9–12 show the data with the template fit at the K^0_S best point of the κ-corrected fits (the Λ panels are therefore a prediction from the K^0_S parameters, not a Λ fit). The uncorrected versions are in Appendix A. Per-bin χ^2: Table 10.

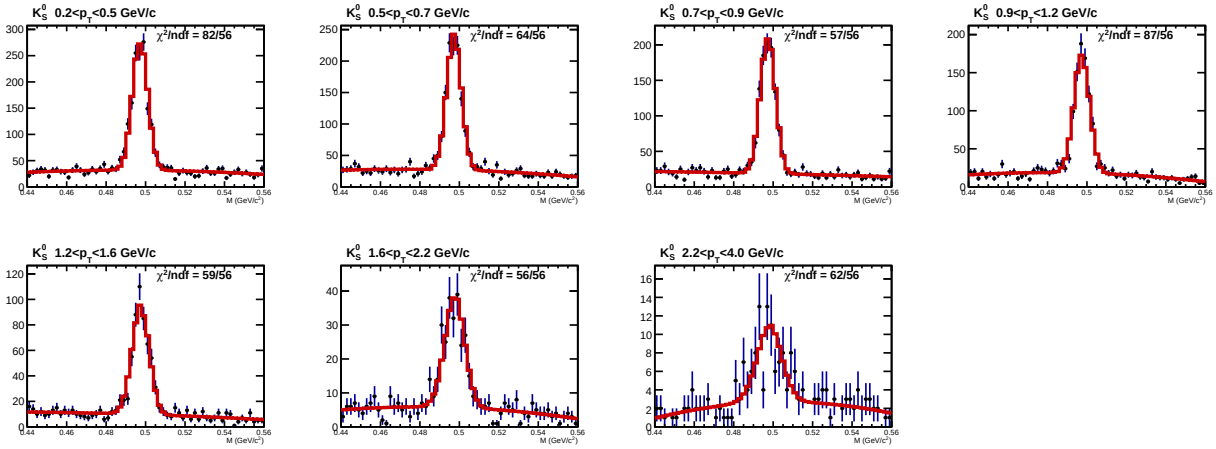


Figure 9: K^0_S mass per V0-p_T bin, period A, with κ (points), and $N\times$DDSim template + quadratic background at the K^0_S best point (red); per-panel Baker–Cousins χ^2/ndf.

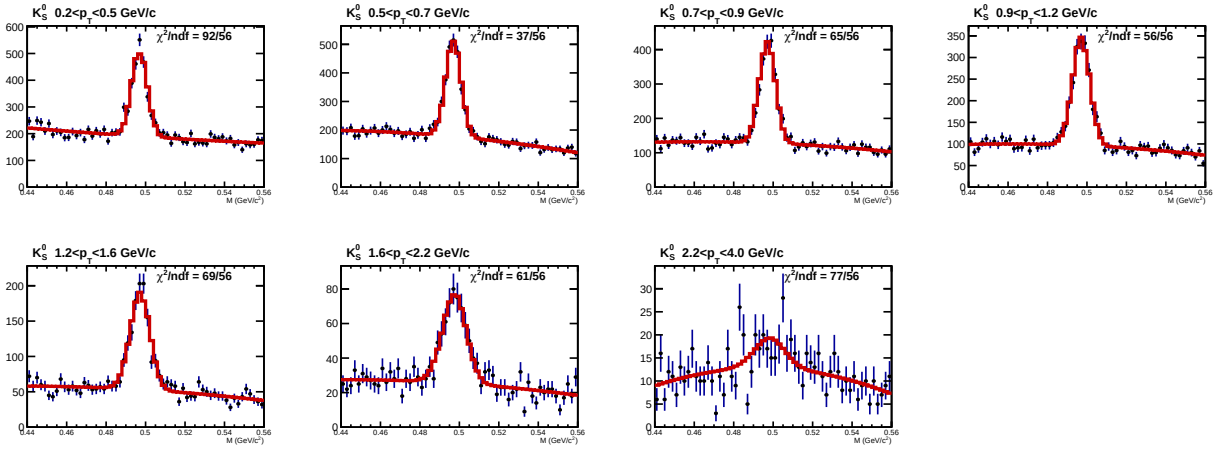


Figure 10: As Fig. 9, period B.

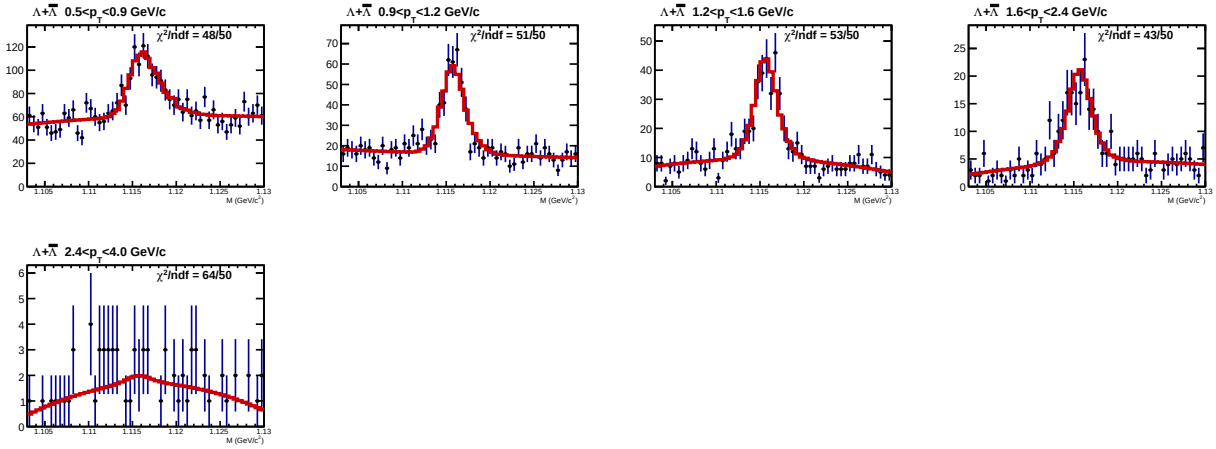


Figure 11: Λ+$\bar{\Lambda}$ mass, period A, with κ; template at the K^0_S best point (prediction).

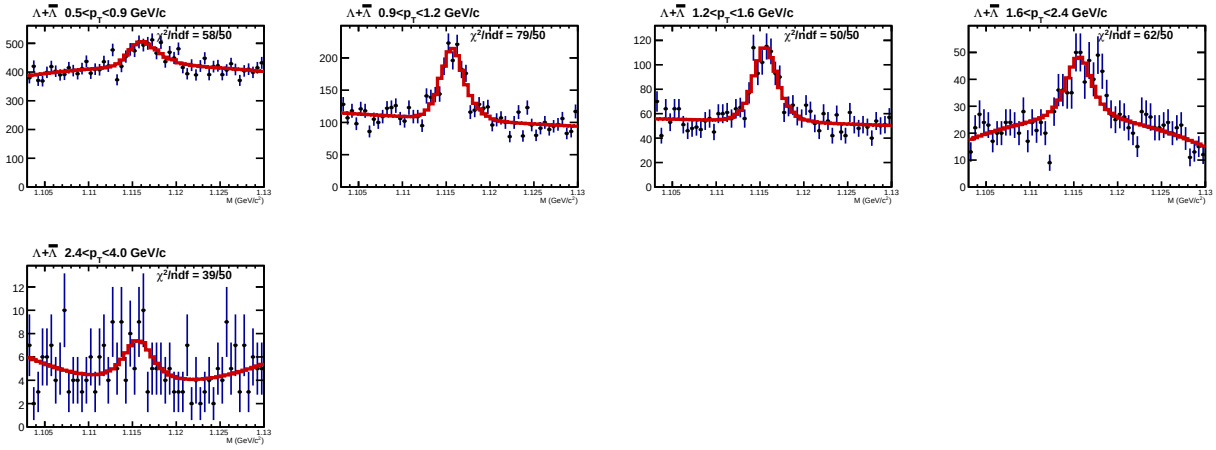


Figure 12: As Fig. 11, period B.

Table 10: Per-bin Baker–Cousins χ^2/ndf of the template fit at the K^0_S best point of each fit, without and with κ.

period		V0 p_T (GeV/c)	no κ	with κ
A	K^0_S	0.2–0.5	45.3/56	82.2/56
A	K^0_S	0.5–0.7	62.4/56	64.1/56
A	K^0_S	0.7–0.9	59.8/56	57.4/56
A	K^0_S	0.9–1.2	53.4/56	87.3/56
A	K^0_S	1.2–1.6	80.6/56	59.5/56
A	K^0_S	1.6–2.2	79.0/56	56.3/56
A	K^0_S	2.2–4.0	84.1/56	61.6/56
A	Λ	0.5–0.9	55.0/50	47.9/50
A	Λ	0.9–1.2	53.6/50	50.7/50
A	Λ	1.2–1.6	54.2/50	53.0/50
A	Λ	1.6–2.4	70.5/50	43.4/50
A	Λ	2.4–4.0	63.2/50	63.7/50
B	K^0_S	0.2–0.5	75.9/56	91.9/56
B	K^0_S	0.5–0.7	58.6/56	36.8/56
B	K^0_S	0.7–0.9	57.0/56	65.4/56
B	K^0_S	0.9–1.2	66.7/56	56.4/56
B	K^0_S	1.2–1.6	58.2/56	69.1/56
B	K^0_S	1.6–2.2	96.4/56	61.1/56
B	K^0_S	2.2–4.0	60.4/56	76.7/56
B	Λ	0.5–0.9	52.1/50	58.0/50
B	Λ	0.9–1.2	80.2/50	78.7/50
B	Λ	1.2–1.6	60.7/50	50.3/50
B	Λ	1.6–2.4	67.6/50	62.4/50
B	Λ	2.4–4.0	57.9/50	39.4/50

8 Tighter Λ selection

The Λ peak sits on a large combinatorial background (S/B ≈ 0.3 within $\pm 2\sigma$ with the QA selection), which limits the precision of its position and makes the Λ template fits sensitive to the background model. This section tests a tighter Λ topological selection, applied identically to data and DDSim templates. K^0_S is not changed.

8.1 Choice of the cuts

Scan. On run 79509 segments 0–30 (full KFParticle ntuple, KFP mass, $0.5 < p_T < 4\,\mathrm{GeV}/c$; `macros/scanLambdaCuts.C`) 720 combinations of DIRA (0.995–0.9999), L_{xy} (0.5–3 cm), pion and proton $|\mathrm{DCA}_{xy}|$ (0.02–0.3 and 0.02–0.1 cm) were evaluated on top of the QA selection. Signal S = counts in $\pm 6\,\mathrm{MeV}/c^2$ around the PDG mass minus a linear side-band estimate (1.095–1.104 and 1.128–1.137 GeV/c^2), B = that estimate; figure of merit $S^2/(S+B)$, the effective signal count that sets the error of the peak position. Only cuts that DDSim models are scanned (no covariance, χ^2 or significance cuts, Sec. 4).

- QA selection: $S = 363$, $B = 1046$, S/B 0.35, $S^2/(S+B) = 94$.
- DIRA is the strongest handle: DIRA>0.9995 alone gives S/B 0.86 and $S^2/(S+B) = 154$ (91% of the signal).
- A tighter decay length removes signal faster than background: $L_{xy} > 2$ cm keeps 26% of the signal (S/B 0.10), because the V0 daughters need silicon hits and therefore decays at small radius.
- Chosen (**tight**): QA + DIRA>0.9995, $|\mathrm{DCA}_{xy}(\pi)| > 0.10$ cm, $|\mathrm{DCA}_{xy}(p)| > 0.03$ cm: $S = 289$ (80%), $B = 168$, S/B 1.72, $S^2/(S+B) = 183$ ($\times 1.96$). The scan maximum (DIRA>0.9997,

$|\mathrm{DCA}_{xy}(\pi)| > 0.15$, $|\mathrm{DCA}_{xy}(p)| > 0.04$ cm, $S^2/(S+B) = 189$) is within the statistical fluctuation of the scan; the less aggressive DIRA cut was preferred (user decision) because the pointing angle is the variable DDSim models least well (Sec. 11).

Can a tight cut bias the resolution fit? DDSim applies the selection before the momentum smearing of the fit (Sec. 6). If a tight cut preferentially removed badly measured candidates in data, the data peak would narrow while the templates would not, and (a, b) would come out too small. Test on the same 31 segments (`macros/cutBiasTest.C`): Gaussian width with the QA and the tight selection in data (KFP mass) and in DDSim (templates smeared at the DDSim main point $a = 3.75\%/(\mathrm{GeV}/c)$, $b = 1.4\%$, $s = 0.991$), for K^0_S with the same DIRA and pion cuts on both pions and for Λ with the tight selection (Table 11). The width ratio tight/QA agrees between data and DDSim within the statistics (K^0_S p_T-integrated: data 0.958 ± 0.042, DDSim 0.985; Λ: 0.986 ± 0.149 vs 0.979). The test is limited to ≈5% by the 31-segment sample. The DDSim Λ templates keep 194,550 of the candidates (0.76 of the QA templates).

Table 11: Cut-bias test on run 79509 segments 0–30: Gaussian width (MeV/c^2) with the QA and the tight selection in data (KFP mass; error of the ratio = quadrature of the two errors, conservative since the tight sample is a subset) and in DDSim at the main point.

	p_T (GeV/c)	data QA	data tight	data ratio	DDSim QA	DDSim tight	DDSim ratio
K^0_S	0.2–0.7	4.70±0.22	4.23±0.15	0.899±0.052	4.63	4.64	1.002
K^0_S	0.7–1.2	5.45±0.32	5.47±0.24	1.005±0.073	5.87	5.86	0.998
K^0_S	1.2–2.2	5.95±0.63	6.86±0.66	1.152±0.165	8.39	8.23	0.981
K^0_S	2.2–4.0	7.51±1.83	8.18±0.27	1.090±0.269	13.05	12.52	0.959
K^0_S	0.2–4.0	5.15±0.17	4.94±0.13	0.958±0.042	5.68	5.59	0.985
$\Lambda+\bar{\Lambda}$	0.5–1.2	2.06±0.30	1.90±0.17	0.922±0.158	1.77	1.74	0.982
$\Lambda+\bar{\Lambda}$	1.2–4.0	2.09±0.53	2.21±0.28	1.054±0.300	2.22	2.15	0.971
$\Lambda+\bar{\Lambda}$	0.5–4.0	2.02±0.27	1.99±0.15	0.986±0.149	1.90	1.86	0.979

8.2 Production

The data ntuples of 256 of the 287 segments are not kept (Sec. 2), so all segments were reprocessed (condor cluster 481293, `run3pp/macro/kappa/fill_v0_mass_kappa_v2.C`, `macro/run_v0mass_kappa_v2.sh`). The v2 fill macro writes, from the same candidates, all histograms of the first production unchanged, the tight histograms `hLa_tight`, `hLaBar_tight` in all three mass versions, and a slim candidate tree per segment (pre-selection DIRA>0.99, $L_{xy} > 0.2$ cm, $|\mathrm{DCA}_{xy}| > 0.01$ cm; 84 MB for all segments, kept locally in `data/kappa_v2/v0tree`) so that further cut variations need no reprocessing. Checks: the v2 macro reproduces the 18 histograms of the first production bin by bin on run 79509 segments 0–30 before submission, and after the production all v1 histograms of all 287 segments are bin-by-bin identical to production 480714 (PASS). In `fitMomRes.C` the option `laTight=1` reads the tight data histograms and applies the same three cuts to the DDSim Λ templates (`rcCosTheta`, `rcPiDcaXY`, `rcPDcaXY`); the K^0_S part of the fit is unchanged (identical K^0_S results: PASS).

8.3 Mass peak

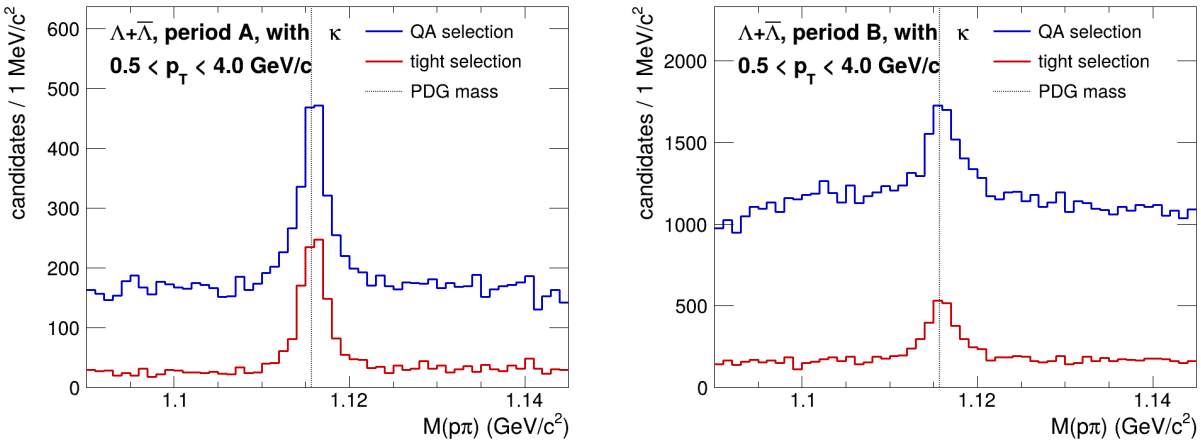


Figure 13: Λ+$\bar{\Lambda}$ mass, recomputed with κ, $0.5 < p_T < 4\,\mathrm{GeV}/c$, with the QA (blue) and the tight (red) selection; period A (left) and B (right). Script `scripts/latight.C`.

Table 12: Λ+$\bar{\Lambda}$ Gaussian peak (gaus + quadratic, 1.103–1.130 GeV/c^2, 0.5 MeV/c^2 bins, likelihood fit), $0.5 < p_T < 4\,\mathrm{GeV}/c$, QA vs tight selection: mean and width (MeV/c^2), S/B within $\pm 2\sigma$, and the fraction of the Gaussian signal kept by the tight cuts. PDG 1115.683 MeV/c^2. `macros/laTightPeaks.C`.

	mass	M QA	M tight	σ QA	σ tight	S/B QA	S/B tight	kept (%)
A	no κ	1115.55±0.12	1115.65±0.10	2.16	2.13	0.81	3.20	71
	with κ	1115.92±0.09	1115.92±0.07	1.63	1.55	1.02	4.19	71
B	no κ	1116.24±0.17	1115.93±0.12	2.70	2.53	0.21	0.87	57
	with κ	1116.11±0.11	1115.96±0.08	1.92	1.79	0.28	1.20	60
all	no κ	1115.99±0.12	1115.81±0.08	2.55	2.35	0.29	1.22	61
	with κ	1116.02±0.08	1115.95±0.06	1.58	1.69	0.35	1.65	74

With κ, the tight selection raises S/B from 0.35 to 1.65 (all segments), keeps 74% of the signal (A 71%, B 60%) and reduces the error of the peak position from 0.076 to 0.056 MeV/c^2 (A 0.087 → 0.071, B 0.108 → 0.080). The peak stays above the PDG mass: 1115.95 ± 0.056 MeV/c^2 (0.26 MeV/c^2, $+4.7\sigma$ statistical).

8.4 Template fits

Table 13: Λ-only template fit with the QA and the tight Λ selection, per period and mass version: profile $\Delta\chi^2 = 1$ intervals, χ^2/ndf, $\Delta\chi^2_\Lambda$ (Λ χ^2 at the K^0_S best point minus its minimum), and the joint K^0_S+Λ best point (a in %/(GeV/c), b in %, s). No best point on the grid edge (PASS).

	mass	Λ sel.	a (%/(GeV/c))	b (%)	s	χ^2/ndf	$\Delta\chi^2_\Lambda$	joint a, b, s
A	no κ	QA	$4.25^{+0.30}_{-0.18}$	1.60	$0.9910^{+0.0007}_{-0.0009}$	293.2/247	3.4	3.00, 1.60, 0.991
		tight	$4.75^{+0.16}_{-0.18}$	1.40	$0.9930^{+0.0015}_{-0.0012}$	244.6/247	4.2	3.00, 1.60, 0.991
	with κ	QA	$3.25^{+0.60}_{-0.45}$	1.20	$0.9980^{+0.0022}_{-0.0004}$	252.7/247	6.1	1.50, 1.40, 0.999
		tight	$2.25^{+0.23}_{-0.33}$	1.20	$0.9990^{+0.0014}_{-0.0004}$	296.6/247	3.4	1.50, 1.40, 0.999
B	no κ	QA	$5.25^{+0.33}_{-0.29}$	3.00	$1.0050^{+0.0009}_{-0.0017}$	287.9/247	30.5	4.00, 1.20, 0.991
		tight	$7.25^{+0.75}_{-0.16}$	1.40	$0.9970^{+0.0021}_{-0.0021}$	288.1/247	38.3	3.75, 1.40, 0.991
	with κ	QA	$3.50^{+0.16}_{-0.72}$	1.40	$1.0010^{+0.0008}_{-0.0014}$	284.0/247	4.8	2.00, 1.40, 0.998
		tight	$4.25^{+0.52}_{-0.21}$	1.20	$1.0000^{+0.0011}_{-0.0014}$	323.9/247	11.9	2.00, 1.40, 0.998
all	no κ	QA	$5.25^{+0.52}_{-0.29}$	2.00	$0.9960^{+0.0025}_{-0.0013}$	280.5/247	25.0	3.50, 1.40, 0.991
		tight	$7.75^{+0.14}_{-0.25}$	1.00	$0.9950^{+0.0011}_{-0.0006}$	263.5/247	45.2	3.50, 1.40, 0.991
	with κ	QA	$2.75^{+0.48}_{-0.63}$	1.20	$1.0000^{+0.0013}_{-0.0011}$	255.7/247	9.0	1.75, 1.40, 0.999
		tight	$4.25^{+0.22}_{-0.31}$	1.00	$0.9990^{+0.0011}_{-0.0003}$	309.7/247	17.3	1.75, 1.40, 0.999

- **Scale.** With κ and the tight selection the Λ-only scale is $s = 0.999$ (A), 1.000 (B) and 0.999 (all), consistent with the K^0_S values (0.999, 0.998, 0.998). The half-width of the $\Delta\chi^2 = 1$ interval in s changes from 1.3 to 0.9 (A), 1.1 to 1.3 (B) and 1.2 to 0.7 (all), in units of 10^{-3}.
- **Resolution, period A.** The tight Λ-only curvature term moves towards the K^0_S value ($a = 3.25 \to 2.25$ vs K^0_S 1.50%/(GeV/c)); $\Delta\chi^2_\Lambda = 6.1 \to 3.4$.
- **Resolution, period B.** The tight Λ-only a rises ($3.50 \to 4.25$) and the Λ/K^0_S tension grows ($\Delta\chi^2_\Lambda = 4.8 \to 11.9$; all segments $9.0 \to 17.3$); the Λ χ^2 worsens ($284.0 \to 323.9$/247). Table 14: at 1.2–1.6 and 1.6–2.4 GeV/c the tight data peaks are wider than the tight templates at the K^0_S best point by 1.4 and 2.0 standard deviations of the data width (QA selection: -0.4 and +0.9).
- With κ the joint K^0_S+Λ best points are unchanged by the tighter Λ selection (K^0_S dominates); without κ the period-B joint point moves by one grid step in a and b.

Table 14: Λ Gaussian width in data / in the template at the K^0_S best point (MeV/c^2), per V0-p_T bin, with κ, QA vs tight selection (from the template-fit display fits, `BIN` rows of the fit logs; "–" = data fit unconstrained or at its 0.3 MeV/c^2 lower limit).

	sel.	0.5–0.9	0.9–1.2	1.2–1.6	1.6–2.4	2.4–4.0
A	QA	2.01±0.24 / 1.70	1.04±0.10 / 1.48	1.73±0.17 / 1.45	1.98±0.24 / 1.57	– / 1.91
	tight	1.63±0.14 / 1.68	1.10±0.07 / 1.44	1.47±0.14 / 1.43	1.41±0.19 / 1.52	– / 1.84
B	QA	2.24±0.32 / 1.74	1.39±0.13 / 1.53	1.49±0.17 / 1.56	1.94±0.19 / 1.77	2.08±0.85 / 2.29
	tight	– / 1.70	1.32±0.10 / 1.50	1.73±0.15 / 1.53	2.11±0.20 / 1.72	– / 2.19

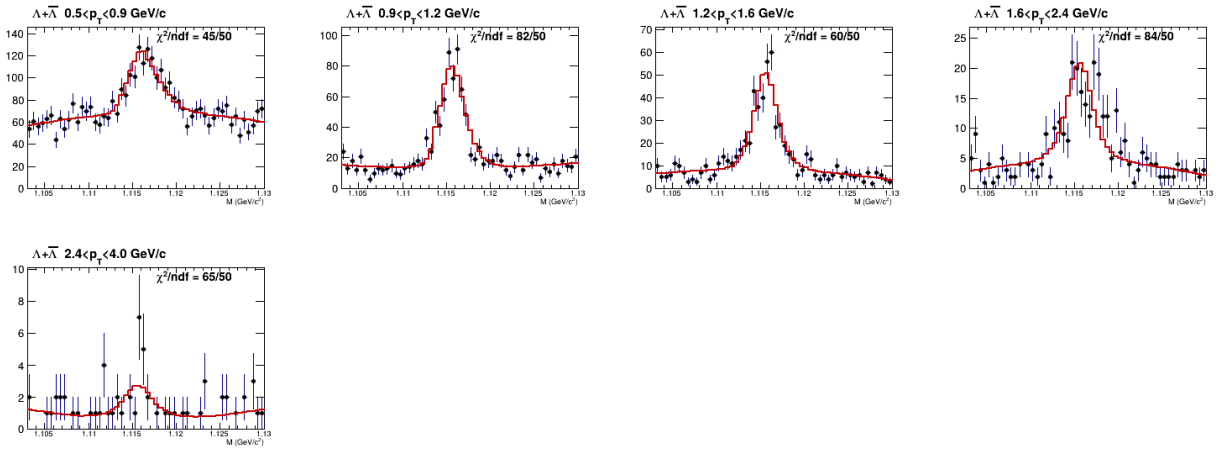


Figure 14: $\Lambda+\bar{\Lambda}$ mass, period B, with κ and the tight selection; template at the K^0_S best point (prediction). Compare Fig. 12 (QA selection).

Choice

Use of the tight selection (adopted): the tight Λ selection is used for the Λ peak position and the Λ momentum scale (s_Λ with κ: 0.999, 1.000, 0.999 for A, B, all). The resolution (a, b) stays determined by K^0_S (Table 6); the Λ resolution with the tight selection is not used until the period-B width mismatch is understood (Sec. 11).

Caveat

Why the tight templates under-predict the period-B width above 1.2 GeV/c is not established. Candidates: the pointing angle is too good in DDSim (true PV, unsmeared directions; in the full-simulation closure study 5% of the true K^0_S have DIRA<0.995 against 0.1% in DDSim), so a DIRA>0.9995 cut selects different decay topologies in data and DDSim; higher pile-up in period B; statistics (the 2.4–4 GeV/c bin has ≈10 tight signal candidates in B). The 31-segment bias test (period A only) shows no effect at the 5% level.

9 Run dependence

The period split was motivated by the observation that the fitted curvature term differs between subsets of the sample. Fits with the KFP mass (no κ) on run groups (Table 15) and per-run counts (Table 16) show:

- a grows with the run period: 2.00%/(GeV/c) on run 79509 segments 0–30, 3.00 on period A, 3.75 on runs 80904–80915 and 4.25 on runs 81556–81609;
- the number of K^0_S QA candidates per event (signal plus combinatorial background) grows in the same order: 0.169–0.231 per run in period A and 0.251–0.980 in period B. This points to higher luminosity or pile-up in the later runs; the luminosity itself is not checked here.

A luminosity-dependent effect such as TPC space-charge distortions would produce this pattern, but this note does not test the hypothesis. The κ table, derived in period A, removes the same fraction of a in both periods (Sec. 7).

Table 15: K^0_S fit with the KFP mass on run groups (same method; s grid 0.975–1.015). Statistical intervals only. Logs `logs/fitMomRes_{kfp31,g7951x,g809xx,g81xxx,k287_*_kfp}.txt`.

runs	a (%/(GeV/c))	b (%)	s	χ^2/ndf
79509 seg. 0–30	$2.00^{+0.76}_{-0.26}$	$1.80^{+0.07}_{-0.24}$	0.990	427.2/389
79509–79516 (A)	$3.00^{+0.06}_{-0.27}$	$1.60^{+0.03}_{-0.05}$	0.991	470.4/389
79510–79516	$3.00^{+0.13}_{-0.15}$	$1.60^{+0.02}_{-0.16}$	0.991	510.2/389
80904–80915	$3.75^{+0.18}_{-0.34}$	$1.20^{+0.10}_{-0.10}$	0.991	395.6/389
81556–81609	$4.25^{+0.79}_{-0.31}$	$1.40^{+0.10}_{-0.48}$	0.990	427.4/389
B (all non-A)	$4.00^{+0.29}_{-0.18}$	$1.20^{+0.15}_{-0.05}$	0.991	400.5/389
all	$3.75^{+0.07}_{-0.27}$	$1.40^{+0.05}_{-0.05}$	0.991	421.5/389

Table 16: Per run: period, segments, processed events, K^0_S QA candidates (all masses in 0.40–0.60 GeV/c^2, signal + background) and per event, Λ+$\bar{\Lambda}$ QA candidates, and mean κ of the QA candidates (K^0_S π^-, π^+; Λ pion, proton). From the condor job logs (`logs/joblines.txt`).

run	per.	seg.	events	K^0_S cand.	K^0_S/ev	Λ cand.	κ_{π^-}	κ_{π^+}	κ^{Λ}_{π}	κ^{Λ}_{p}
79509	A	39	38,656	8,939	0.231	17,656	0.9980	1.0164	1.0078	1.0067
79510	A	8	7,931	1,599	0.202	3,306	0.9978	1.0160	1.0077	1.0067
79511	A	8	7,934	1,717	0.216	3,324	0.9980	1.0160	1.0076	1.0063
79512	A	8	7,932	1,663	0.210	3,087	0.9969	1.0167	1.0076	1.0063
79513	A	8	7,931	1,619	0.204	2,942	0.9975	1.0164	1.0079	1.0063
79514	A	8	7,931	1,493	0.188	2,783	0.9977	1.0161	1.0076	1.0063
79515	A	8	7,930	1,473	0.186	2,623	0.9974	1.0160	1.0079	1.0052
79516	A	8	7,928	1,338	0.169	2,379	0.9987	1.0162	1.0076	1.0072
80904	B	8	7,936	3,778	0.476	8,545	0.9971	1.0160	1.0076	1.0063
80905	B	8	7,938	3,572	0.450	8,029	0.9974	1.0162	1.0078	1.0064
80906	B	8	7,938	3,601	0.454	7,882	0.9974	1.0160	1.0077	1.0066
80908	B	8	7,939	3,036	0.382	6,692	0.9983	1.0158	1.0075	1.0066
80909	B	8	7,936	2,666	0.336	6,117	0.9968	1.0162	1.0077	1.0058
80913	B	8	7,930	2,166	0.273	4,397	0.9971	1.0162	1.0077	1.0060
80915	B	8	7,933	1,994	0.251	3,796	0.9972	1.0161	1.0073	1.0064
81556	B	8	7,936	7,515	0.947	17,241	0.9978	1.0163	1.0076	1.0068
81558	B	8	7,937	7,086	0.893	16,020	0.9981	1.0163	1.0077	1.0070
81560	B	8	7,941	4,724	0.595	10,797	0.9980	1.0157	1.0077	1.0068
81562	B	8	7,939	4,152	0.523	9,505	0.9980	1.0162	1.0077	1.0067
81566	B	8	7,933	3,116	0.393	6,854	0.9975	1.0159	1.0078	1.0062
81575	B	8	7,938	6,779	0.854	16,118	0.9978	1.0165	1.0079	1.0069
81576	B	8	7,936	6,319	0.796	14,385	0.9977	1.0162	1.0077	1.0068
81577	B	8	7,937	6,031	0.760	13,923	0.9979	1.0165	1.0077	1.0069
81580	B	8	7,940	4,564	0.575	10,359	0.9978	1.0156	1.0076	1.0066
81581	B	8	7,940	4,384	0.552	9,782	0.9979	1.0165	1.0077	1.0069
81583	B	8	7,937	3,930	0.495	8,828	0.9977	1.0165	1.0079	1.0064
81584	B	8	7,939	3,446	0.434	7,746	0.9978	1.0160	1.0078	1.0061
81585	B	8	7,933	3,335	0.420	7,403	0.9983	1.0159	1.0078	1.0067
81586	B	8	7,931	3,403	0.429	7,605	0.9981	1.0159	1.0079	1.0065
81600	B	8	7,935	7,773	0.980	18,212	0.9974	1.0163	1.0078	1.0066
81601	B	8	7,937	6,193	0.780	14,406	0.9977	1.0161	1.0076	1.0067
81609	B	8	7,936	3,855	0.486	8,527	0.9972	1.0160	1.0078	1.0064

10 Audit

Checks that passed (script-computed, `scripts/make_numbers.py`).

- Jobs: 287/287 condor jobs printed "OK" with three non-empty outputs (`logs/joblines.txt`): PASS.
- Reproducibility: in all 287 segments the KFP-mass histograms of this production (`hKs_qa`, `hKs_loose`, Λ+$\bar{\Lambda}$ QA) are bin-by-bin identical to the earlier covariance-free production 480608 (287/287; `run3pp/output/v0mass_kappa_validate/cmpKfpRepro.C`): PASS.
- Fill macro: the cluster macro `fill_v0_mass_kappa.C`, run on the stored run 79509 segment 0–30 ntuples, gives bin-by-bin the same 18 histograms (3 versions × 6) as the independent

local implementation `macros/fillV0MassCorr.C` (`macros/cmpKappaFill.C`): PASS.

- κ table complete (72,800 = 2×112×13×25 nodes, none missing): PASS; interpolation at table nodes returns the node value ($q = \pm 1$, $p_T = 1\,\mathrm{GeV}/c$, $\eta = \varphi = 0$): PASS.
- Tight-Λ production 481293: all v1 histograms of the 287 segments bin-by-bin identical to 480714: PASS; K^0_S fit results identical with and without the tight Λ option: PASS; no tight-fit best point on the grid edge: PASS.
- No K^0_S or joint best point on the grid boundary in the fits of Table 7: PASS.

Not verified (assumptions).

- That the KFParticle secondary-vertex-constrained daughter momenta are the kinematics the κ table was trained on.
- That the table holds outside runs 79509–79516 (period B); the improvement there is the only evidence.
- The effect of keeping the uncorrected DIRA, L_{xy} and DCA values in the selection (the correction changes the V0 direction slightly).
- Double counting of the proton scale: the templates contain the full-simulation proton bias δ_p, and κ corrects the measured proton momentum as well.
- The DCA_{12} cut of the templates (0.2 cm unconstrained) vs the effective data cut (Sec. 4).
- That all V0 daughters are silicon-matched (seen in the earlier run3pp topology study through the crossing requirement; not rechecked on this production, which keeps only histograms).
- Background shape: quadratic in every bin; no variation in this note (the DDSim calibration found the Λ result sensitive to it).

11 Open issues

1. [decision] Adoption of the correction in DDSim. If the κ-corrected response is the target, DDSim needs per-period (a, b, s) (Table 6, "with κ") and the analyses must apply κ to their data tracks; the current DDSim defaults ($a = 3.75\%/(\mathrm{GeV}/c)$, $b = 1.40\%$, $s = 0.991$) describe uncorrected data averaged over all runs.
2. Period B keeps a larger curvature term after the correction (2.00 vs $1.50\%/(\mathrm{GeV}/c)$). A table derived on period-B runs, or a per-run fit, would show whether the remainder is a run-dependent distortion that κ does not describe.
3. KFP mass vs recomputed mass: in period B the K^0_S χ^2 is 400.5 with the KFP mass and 473.1 with the recomputed mass (best points $(a, b, s) = (4.00\%/(\mathrm{GeV}/c), 1.20\%, 0.991)$ and (4.00, 1.20, 0.991)); in period A the two are similar (470.4 vs 464.5). The cause (vertex-fit treatment, background shape of the recomputed spectrum) is not understood.
4. Residual K^0_S peak offset with κ: 0.40 (A) and $0.57\,\mathrm{MeV}/c^2$ (B) below PDG; fitted $s = 0.999$ and 0.998.
5. Proton scale: check whether δ_p in the templates should be removed when κ is applied (Λ-only s with κ: 0.998 and 1.001).
6. Tight Λ selection in period B: the tight data peaks above $1.2\,\mathrm{GeV}/c$ are wider than the tight templates ($\Delta\chi^2_\Lambda$ 4.8 → 11.9). Next step (local, slim trees): refill with DIRA alone and with the DCA cuts alone to locate the cause, and compare the DIRA distribution of side-band-subtracted Λ in data and DDSim.

7. Λ peak position with κ and tight selection $0.26\,\mathrm{MeV}/c^2$ above PDG ($+4.7\sigma$ statistical) while K^0_S is 0.40 (A) and $0.57\,\mathrm{MeV}/c^2$ (B) below: the proton scale (item 5) is the first suspect.
8. Background-model and window variations, as in the DDSim main calibration, have not been redone with the corrected masses.
9. Regenerate the templates with the current DDSim defaults ($\mathrm{DCA}_{12} < 0.5$ cm, seeding factor) and refit.

Decision needed

Which response should DDSim use? Options: (a) keep the uncorrected all-run defaults; (b) adopt the κ-corrected response per period (requires the analyses to correct their data with κ); (c) wait for a κ table covering period B.

12 Glossary

κ per-track multiplier of the transverse momentum from the look-up table, $p_T^{\mathrm{corr}} = \kappa p_T$.

a, b, s curvature term, multiple-scattering term and global scale of the momentum response, Eq. 1.

period A / B runs 79509–79516 (range of the table) / all other runs of the sample.

KFP / no κ / with κ the three mass versions: KFParticle fitted mass; recomputed from the stored daughter momenta; recomputed after the correction.

DDSim data-driven fast simulation of two-body V0 decays (DataDrivenSim project).

DIRA cosine of the angle between the V0 momentum and the PV-to-SV flight direction.

DCA_{12}, DCA_{xy} distance of closest approach between the daughters; transverse DCA of a daughter to the PV.

L_{xy} transverse decay length (PV to SV).

QA cuts offline V0 selection of Table 2.

tight Λ selection QA + DIRA>0.9995, $|\mathrm{DCA}_{xy}(\pi)| > 0.10$, $|\mathrm{DCA}_{xy}(p)| > 0.03$ cm (Sec. 8).

Baker–Cousins χ^2 $2\sum_i[\mu_i - d_i + d_i \ln(d_i/\mu_i)]$, the Poisson likelihood-ratio goodness of fit.

segment one DST file of a run, about 990 events.

A Template fits without the correction

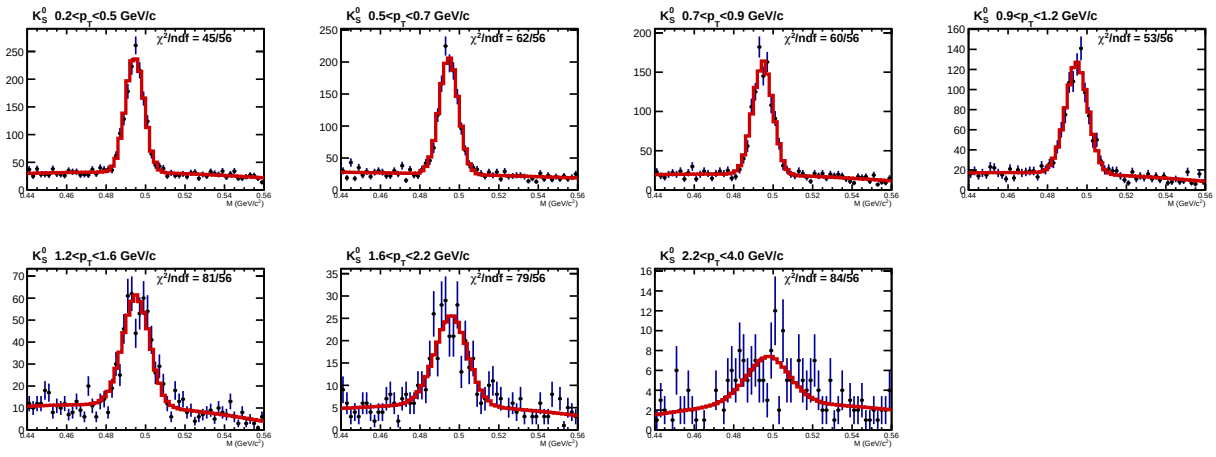


Figure 15: K^0_S mass per V0-p_T bin, period A, recomputed without κ, with the template fit at that fit's K^0_S best point (red).

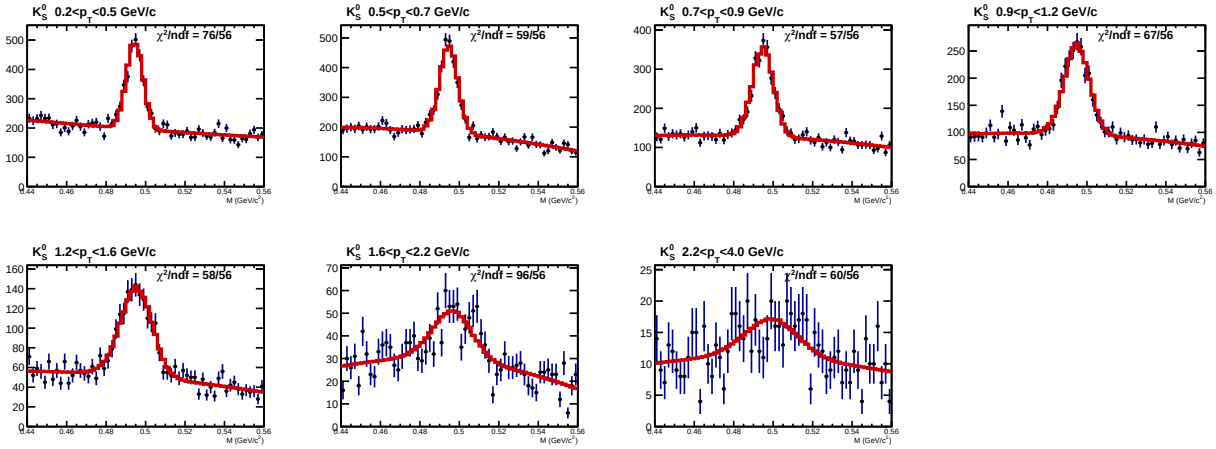


Figure 16: As above, period B.

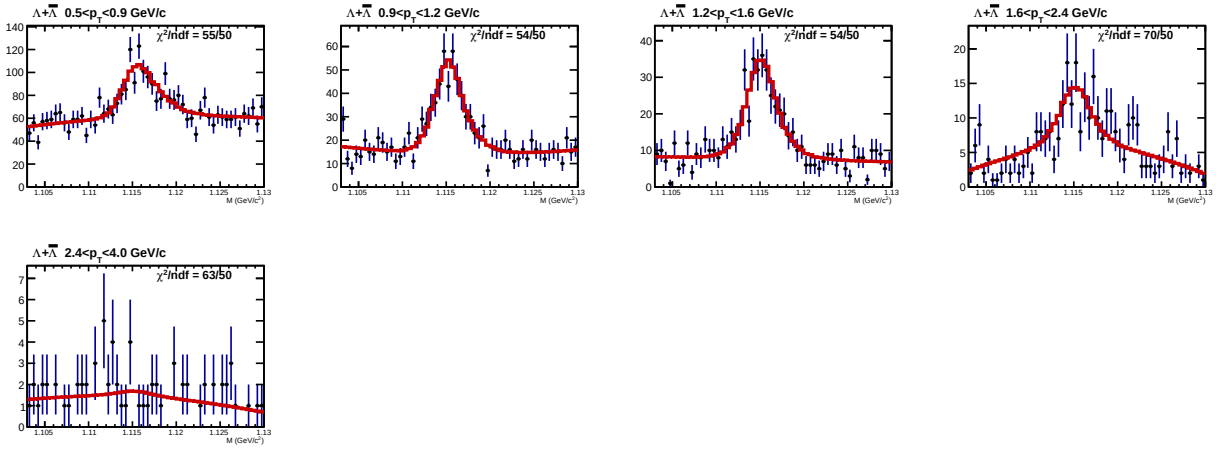


Figure 17: Λ+$\bar{\Lambda}$, period A, without κ; template at the K^0_S best point of the same fit.

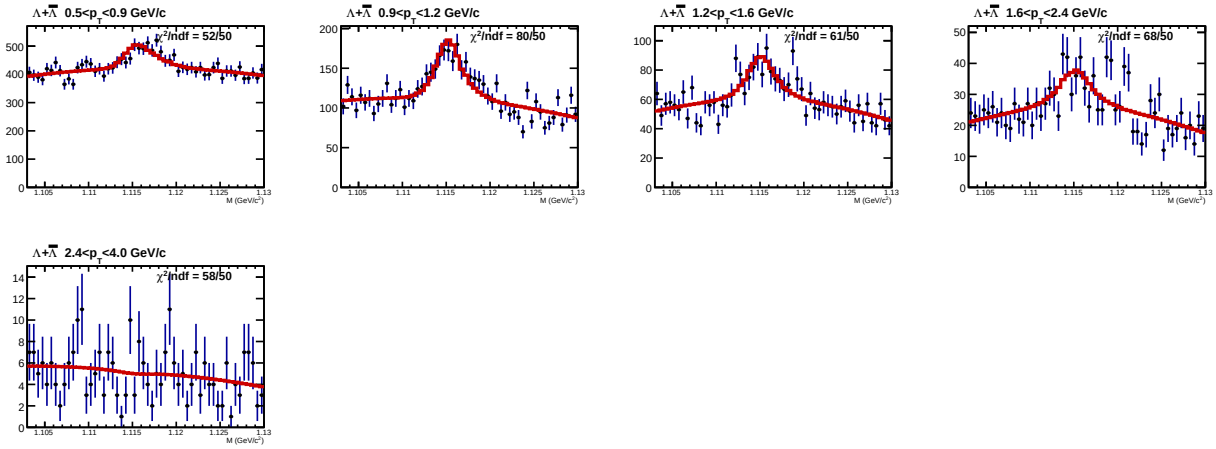


Figure 18: As above, period B.